

Agents Propose, Code Decides: An Autonomous Research Loop Framework to Build an Equity Alpha Model

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October 2026

ABSTRACT

This paper proposes an autonomous research loop in which large language model (LLM) agents build a multi-factor equity alpha model from start to finish, while a fixed, rule-based harness keeps every step auditable. Agents decide what to test, and code decides how it is measured. A runner model (Claude Opus) executes the search from a standing instruction file, an advisor model (Claude Fable) reviews phase boundaries and acceptances, and five sub-agents with restricted tools fetch, verify, translate, audit and evaluate each predictor in Chen and Zimmermann's (2022) Open Source Asset Pricing catalogue on a frozen Sharadar dataset. The acceptance bars, a long-short ranked within sector and hedged to the market by an ex-ante beta, and a 57-month out-of-sample period were fixed before any candidate was screened, and the human answers a closed list of seven questions rather than directing the search.

The loop inventoried all 212 catalogue predictors in Chen and Zimmermann (2022), screened the 107 the data could construct, and admitted 14 whose information was significant beyond every leg already held, raising a Fama-French (2015) plus-momentum baseline's mean rank IC from 0.0145 to 0.0389 over 1999 to 2021. Out of sample the IC held at 0.0300, while the hedged long-short earned about half its expected Sharpe (0.523 against 0.996), most of it from the hedge term. The contribution is the framework rather than the model: a research process that runs itself, traces every number to the data and code that produced it, and reports a mixed result against expectations it wrote down before it looked.

JEL classification: G11, G12, G14, C58, C88

Keywords: autonomous research loop, AI agents, large language models, agentic workflows, factor investing, alpha model, pre-registration, market-neutral construction, long-short strategies, liquid alternatives

*Erfan Sadeghi is an independent researcher (sadeghi1992@gmail.com). The code, the records and every result are at <https://github.com/Erfan94/autonomous-factor-research>. The search was specified by the author and executed by Claude large language model agents (Anthropic). The bootstrap session ran on Claude Fable 5.1; the search ran on Claude Opus 5.5 with Claude Fable 5.1 as advisor. The author is responsible for the content. The author has no financial interest in any security discussed.

I. Introduction

Large language model (LLM) agents can now read a paper's code, plan a piece of research, execute it and interpret what comes out. That makes it possible to hand an agent a research programme rather than a single task. In cross-sectional equity research the natural programme is the factor search: take a catalogue of published predictors, rebuild each one on your universe, test it, and assemble the survivors into an alpha model. An agent can carry out every step of that search, and it can do so faster and more exhaustively than a person. The question this paper asks is what it takes for the result to be trustworthy.

The difficulty is that the factor search is exactly where empirical finance goes wrong. Harvey, Liu, and Zhu (2016) counted more than three hundred published predictors and argued that a t-statistic of two no longer carries its textbook meaning once that many have been tried. McLean and Pontiff (2016) showed that predictor returns fall by about a third after publication, and Hou, Xue, and Zhang (2020) found that most published anomalies do not survive a careful replication. Chen and Zimmermann (2022) answered by releasing open, reproducible code for the whole catalogue, the Open Source Asset Pricing (OSAP) project. That removed one source of error and left the search itself as the main one. The failures are procedural. The number of predictors tried is rarely written down. Thresholds, and the order in which candidates are tested, tend to settle after the results are in. The backtest code changes between the first candidate and the fortieth, and neither result says so. The holdout is looked at more than once. And a predictor that is strong on its own but redundant with the model already held is added anyway, bringing noise and turnover rather than information.

Handing the search to an agent does not remove these failures; it makes them worse. An agent can test thousands of variants in an afternoon, far more than any person, and unless something records them it will not remember in its next session what it already tried. It can change the code that measures results as easily as the code that defines a factor, and it can misread its own output, as it did in this project (Section XI). A careful researcher guards against

these failures through personal discipline. An agent cannot be relied on to do so, so the protection has to come from the system around it: rules and records that the agent can neither change nor forget.

This paper proposes such an environment, an autonomous research loop, and reports its first complete run. The loop rests on one division of labour: agents propose, code decides. Agents do the open-ended work: reading a reference implementation, mapping its Compustat and CRSP inputs to a vendor's fields, writing the factor definition, auditing it, interpreting a result and writing the record. A fixed, rule-based harness does everything that must be identical across candidates: building the point-in-time universe, ranking within sector, forming deciles, hedging the long-short to the market, computing the statistics and applying the acceptance rules. Around both sits a layer of enforcement written in code, made of phase checkpoints, commit hooks, permission tiers and cryptographic stamps, that no agent can talk its way past. The human designs the rules, supplies the data, and answers seven questions the loop may not decide alone. Everything else, from which predictor to translate next to when to fix the harness and what to write down, the loop decides and logs.

The paper makes four contributions. First, it describes the architecture of a loop in which agents run a factor search end to end without being able to change how results are measured (Section II). Second, it shows that uniform measurement and pre-registration can be made properties of the data and code rather than of anyone's conduct. Four hashes are stamped on every result, the order of the phases is enforced in code, and nothing in the record is edited after the fact. Third, it shows how an agent loop can take a whole catalogue as its denominator: every predictor is inventoried, and every one not tested is written down with a measured reason. Fourth, it pairs the loop with a selection rule and a construction suited to automation. The composite is a blend of economic families, so near-duplicate signals cannot take whole votes, and a signal is admitted only when its information beyond every existing leg is significant. Every signal is ranked within its sector, and every long-short is hedged to the market with a beta estimated on past months only; the unhedged series and its beta are reported alongside. Signals are selected for

the information they add, not for their in-sample returns. How the model would be traded is reported separately and plays no part in selection.

Within three days, the loop inventoried all 212 predictors in the Chen and Zimmermann (2022) catalogue, ran 107 screens over the 106 predictors that Sharadar's data can construct and one declared reversal, ratchet-tested the 24 that passed the standalone screen, and accepted 14. It took a five-leg Fama-French-plus-momentum baseline to a 19-leg composite in 9 families, and the mean rank IC rose from 0.0145 to 0.0389 over 1999 to 2021. It then stopped for my approval, updated its data under a declared protocol and evaluated the reserved 57-month out-of-sample period once. Three results stand out. The first is that the rules held, including where they were known to be imperfect. The hedged return guard carries a disclosed bias in favour of legs that push the composite's market beta down, and the loop reported the bias rather than repairing it with a new bar; every one of the 10 ratchet rejections fell on the information bar, and the guard never bound. The second is that the information bars accumulated a market exposure they do not measure. As the ratchet admitted low-volatility, illiquidity and profitability legs, the full-window beta of the composite's raw long-short went from -0.140 to -0.560 ; the hedge removed that exposure on paper, and every acceptance-time headline is the hedged one. The third is that out of sample the ranking information held but the return did not. Over the out-of-sample months the composite's IC was 0.0300, above the second-half in-sample benchmark named in the expectations written before the out-of-sample evaluation, and significant at the 10% level (t-stat 1.87); the hedged long-short earned 10.81% a year at a Sharpe of 0.523 against an expected 0.996, and 7.79 percentage points of that return were the hedge term, the long market position a negative-beta hedge adds, earned in a rising market. The hedge was also too small: its trailing beta lagged a book whose realised beta was -0.836 , so the hedged series stayed partly short the market.

The number I would defend most strongly is not the IC gain but the ledger around it: 83 predictors rejected at the screen, 10 rejected at the ratchet, 101 excluded for a logged reason, and one holdout evaluated once, with its expectations written first. Every rejection is logged with the

same detail as every acceptance, so a reader can see what the search tried, why each candidate failed, and what the model did on data it had never seen. The mixed out-of-sample result is what the framework produced, and it is the useful result. It was not chosen, it matched what the in-sample diagnostics had already printed about the beta and the regime concentration, and it could be reported without retuning anything because how it would be read was committed in advance.

The paper is organised as follows. Section II describes the framework in detail: its principles, its participants, the instruction file that drives the runner, the phases and their checkpoints, one predictor’s path through the loop, and the mechanisms for provenance, memory, autonomy and review. Section III describes the data: the Sharadar dataset, the predictor catalogue and the monthly universe. Section IV describes what the harness measures, including the within-sector ranks and the market hedge, and Section V sets out the selection rules. Sections VI to IX report the loop’s first complete run as a case study: the search phase by phase, the factors it selected, the construction layer, and the out-of-sample result. Section X reads the result, Section XI states the limitations, and Section XII says which parts of the design transfer. The appendices hold the full ledgers: every ratchet rung, every Stage 1 rejection, every predictor not tested and why, the agents’ division of labour, the reproduction proofs, and a description of each accepted leg.

II. The Framework

A. Design Principles

Six principles shaped every component of the loop. Each answers one of the procedural failures above, and each is enforced by something a reader can inspect.

Agents propose, code decides. An agent may decide what to test. It may never decide how the test is measured. Factor files read data only through a month context that the harness supplies, and they may not contain a universe filter, a date, a rebalance rule, a sector ranking, a hedge or any statistic. A factor that needs different machinery has to extend the harness, which changes the harness hash and makes the change visible on every later result.

Provenance is a property of the data and code. Every result carries hashes of the code, the configuration, the composite and the data that produced it, and a result whose hashes do not match the repository is refused. Nobody has to be trusted to have run the right thing.

Every rule is written before it is needed. The bars, the test order, the Newey-West lag, the out-of-sample boundary, the way the long-short is ranked and hedged, and the expectations against which the out-of-sample period would be read were fixed before the scores they govern existed. Where the loop had to interpret a rule during the search, it logged the ruling and never introduced a new rule.

The denominator is the whole catalogue. The loop does not propose its own candidates. It inventories every predictor in a public catalogue, here the Open Source Asset Pricing catalogue of Chen and Zimmermann (2022), and accounts for each one, either with a test result or with a measured reason why it could not be tested.

Autonomy has a closed boundary. The loop decides everything except a fixed list of seven questions. The list is short enough that the loop can run for days without a human. It is closed, so the loop cannot widen its own authority.

Everything is written down, rejections included. Every run, verdict, ruling and correction goes into an append-only record in the same detail, so the rejection table is as complete as the acceptance table.

B. Architecture and Participants

Figure 1 shows the loop's participants and the layers that constrain them. The orchestration layer decides what to do next. The runner, an LLM working from a standing instruction file, drives the search. It consults a stronger advisor model at fixed points and puts seven kinds of questions to the human. The sub-agent layer does the specialised work that needs judgement. The measurement layer is a fixed, rule-based Python harness that computes every statistic and applies every bar. The record layer holds everything the loop has done. Beneath all four, a layer of scripts, hooks and permission rules enforces the boundaries in code.

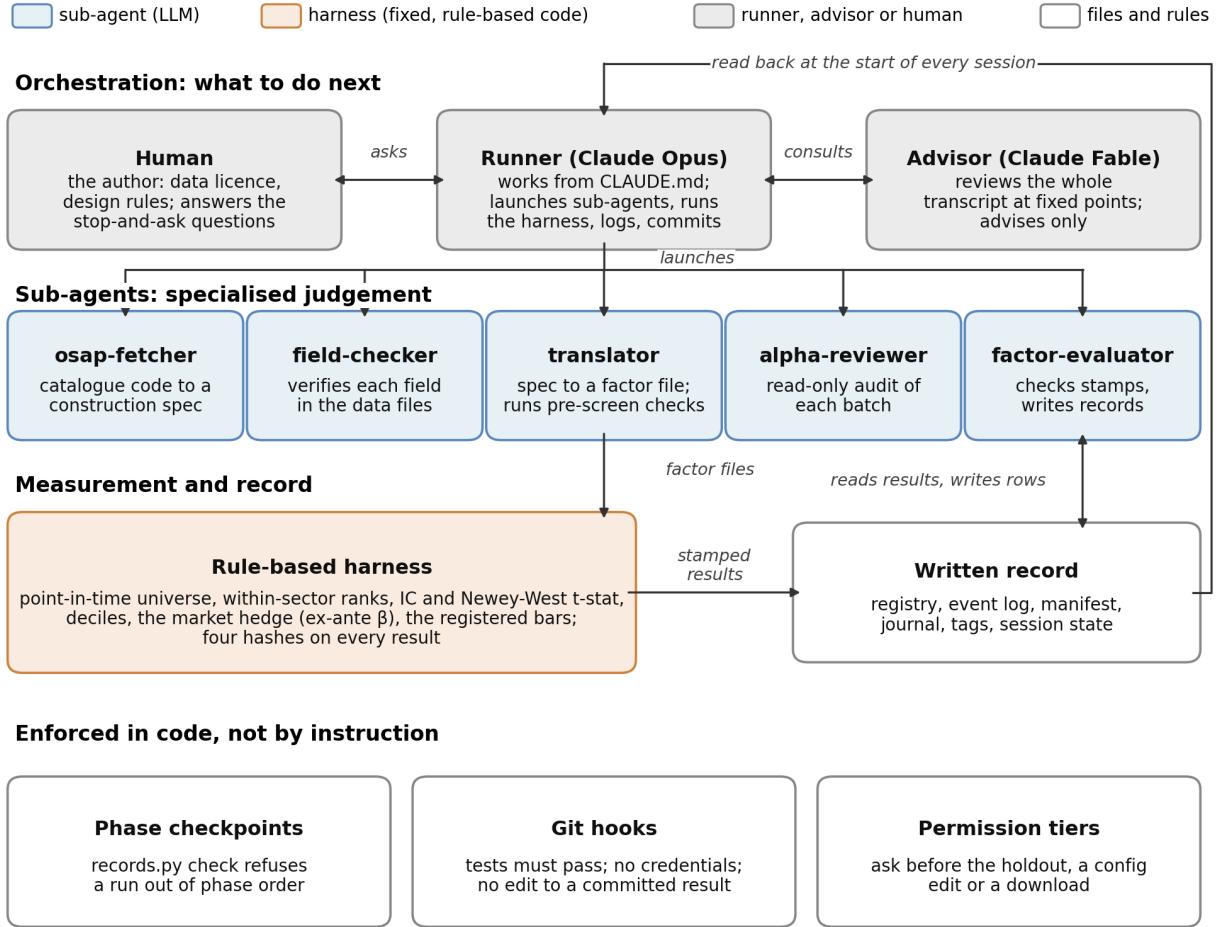


Figure 1. The architecture of the loop. The runner decides what to do next, consults the advisor at fixed points and asks the human only the seven stop-and-ask questions. It launches five sub-agents, each with its own instructions and restricted tools. The translator’s factor files go to the harness, which ranks within sector, hedges every long-short to the market, measures every candidate identically and stamps every result. The evaluator checks those results and writes the record, and the record is read back at the start of the next session, which closes the loop. The bottom row is enforced by scripts, git hooks and permission rules rather than by instruction.

Table I lists the participants. Each sub-agent is defined by its own instruction file, which states its job, its inputs, what it must report and what it must never do, together with a list of the tools it may use. The adversarial reviewer, for example, can read and search files but cannot write or run anything. Its instructions tell it to assume the code is wrong until verified and to treat an unusually strong result as a reason for more suspicion. The fetcher may take reference code only from the catalogue’s pinned commit, and its specification describes construction only; it may not mention how a predictor performed anywhere else. The translator writes one factor file and may not put a universe filter, a date or a statistic into it. Nor may it assign the factor a family, which

happens later and elsewhere. The evaluator checks provenance before anything else and refuses to write a record for a result whose stamps disagree with the repository. Appendix D gives the full division of labour.

Table I
The Participants

Each participant in the loop, the model it runs on, its job and the constraint it works under. Models are those recorded in docs/JOURNAL.md and declared in each sub-agent's definition file (.claude/agents/); “inherits” means the sub-agent runs on the runner’s model.

Participant	Model	Job	Constraint
Human (the author)		Supplies the data licence; fixed the design rules before the loop started; answers the seven stop-and-ask questions.	Never edits a record by hand.
Runner	Claude Opus (Claude Fable at bootstrap)	Runs the loop from the standing instruction file: reads the state, launches sub-agents, runs the harness, logs decisions, commits and tags.	Decides anything not on the stop-and-ask list; may not edit a past result or move a stamp while a run is unevaluated.
Advisor	Claude Fable	Reviews the runner’s whole transcript at phase boundaries, before a ladder’s acceptances are committed, before any stop-and-ask, and when a result looks too good.	Advises only. Advice that changes a decision is logged.
osap-fetcher	Claude Sonnet	One predictor’s reference code and catalogue row at the pinned commit, written up as a construction-only specification.	Pinned commit only; no outcome from anywhere.
sharadar-field-checker	Claude Sonnet	Verifies each field a factor needs by querying the data files: existence, units, null semantics, point-in-time shape.	Read-only queries; one writer of the field map at a time.
sharadar-translator	Claude Sonnet	Turns a specification into one factor file and runs the pre-screen data checks on it.	No universe, dates, rebalancing, sector ranking, hedging or statistics in a factor; family left unset.
alpha-reviewer	inherits the runner’s	Adversarial audit of every translated batch and every harness change for look-ahead, survivorship and point-in-time errors.	Read-only; reports, never fixes.
factor-evaluator	inherits the runner’s	Checks the four stamps, applies the bars, re-derives every verdict, explains the numbers, writes every record, and implements each acceptance.	Refuses a mismatched stamp; its verdict must equal the harness’s.

C. The Standing Instruction File

The runner has no memory between sessions and no goal of its own. Both come from a standing instruction file in the repository’s root, which the runner loads at the start of every session. The file is short, a few pages, and it works as the program the runner executes. It states the goal: a multi-factor model, with every candidate recorded. It states the one constraint that shapes everything, that every test runs on the same stamped data and code. It gives the pre-registered rules in summary, including the construction choices, the seven stop-and-ask

questions, the five phases and what closes each, a list of hard rules, the table of records the loop must keep, and the commands that run each step.

Two instructions in the file do most of the work of making the loop autonomous. The first is “a turn ends with a run executed and a decision logged, not with a question”. When the runner is about to ask the human something that the rules already answer, it is told to apply the rule, say in one sentence which rule it applied, and continue. The second fixes the shape of every reply. Each one ends with a table of every composite version and its headline statistics, a line saying how many candidates have been accepted, screened and inventoried and which phase the search is in, and one sentence saying what is running, what the human owes, or what is blocked. A human who looks in at any moment can see the state of the search without reading the transcript.

The file is written in plain language, and plain language is not enforcement. Every rule in it that can be checked by a script is also checked by one. The file says Stage 2 may not start before the families are declared, and a records checker refuses the run. It says the holdout is evaluated once, and the harness flags any result that extends into 2022 as an out-of-sample breach, while the permission layer asks the human before either option that runs the out-of-sample test can be used. The instruction file tells the runner what to do. The code makes sure the important parts were done.

The instruction file is also the part of the framework that a reader would change to point the loop at a different problem. The catalogue, the data vendor, the bars, the construction and the phases are all named there. The mechanisms underneath, the stamps, the checkpoints, the hooks and the record, do not depend on any of them.

D. The Phase Machine

The search runs in five phases after a one-time setup, and no phase starts before the previous one closes (Table II and Figure 2). The order is itself a safeguard. The loop inventories everything before it screens anything, so it cannot choose what to examine after seeing results. It screens every constructible predictor before it assigns any family, and it assigns every family and

declares the test order before any Stage 2 number exists, so neither the families nor the order can be fitted to marginal results.

Table II
The Phases of the Loop

What each phase does, who does it, what closes it, and what enforces the close.

Phase	Work	Who does it	Closes when	Enforced by
0. Setup	Download and freeze the data, prove they match the vendor's live database, measure the baseline and tag it.	Human and a bootstrap session	The baseline is measured, recorded and tagged before any candidate exists.	The data hash and the baseline tag.
A. Inventory	For every catalogue predictor: fetch, check fields, translate, run the pre-screen data checks; review every batch. Batches of eight, in alphabetical order.	Fetcher, field checker, translator, reviewer	Every predictor has a file that passed the pre-screen checks or a logged, measured reason.	The records checker reconciles the catalogue against files and exclusions; none may be unaccounted.
B. Stage 1	Screen every constructible predictor on its own. Batches of twelve, in alphabetical order.	Harness; evaluator	Every constructible predictor has a Stage 1 row.	The records checker refuses a Stage 2 run while any row is missing.
C. Families	Assign each passer to an economic family by definition; then sort all passers by Stage 1 t-stat and declare the order.	Runner, advisor	The order file is written, before any Stage 2 number.	The harness refuses a Stage 2 candidate without a family or beyond the family cap.
D. Ratchet	Test the passers in declared order, in ladders of at most five; implement every acceptance as a tagged version.	Harness; evaluator; advisor before each commit	Every passer has a Stage 2 row.	The order is never re-sorted; the harness hash may not move inside a ladder.
E. Construction and holdout	Build the construction layer on the frozen composite; update the data; ask to declare the search finished; evaluate the holdout once; write the paper.	Runner, advisor, human	The holdout is evaluated.	A result extending into 2022 is an out-of-sample breach unless it is the final validation; the permission layer asks before the out-of-sample test is run.

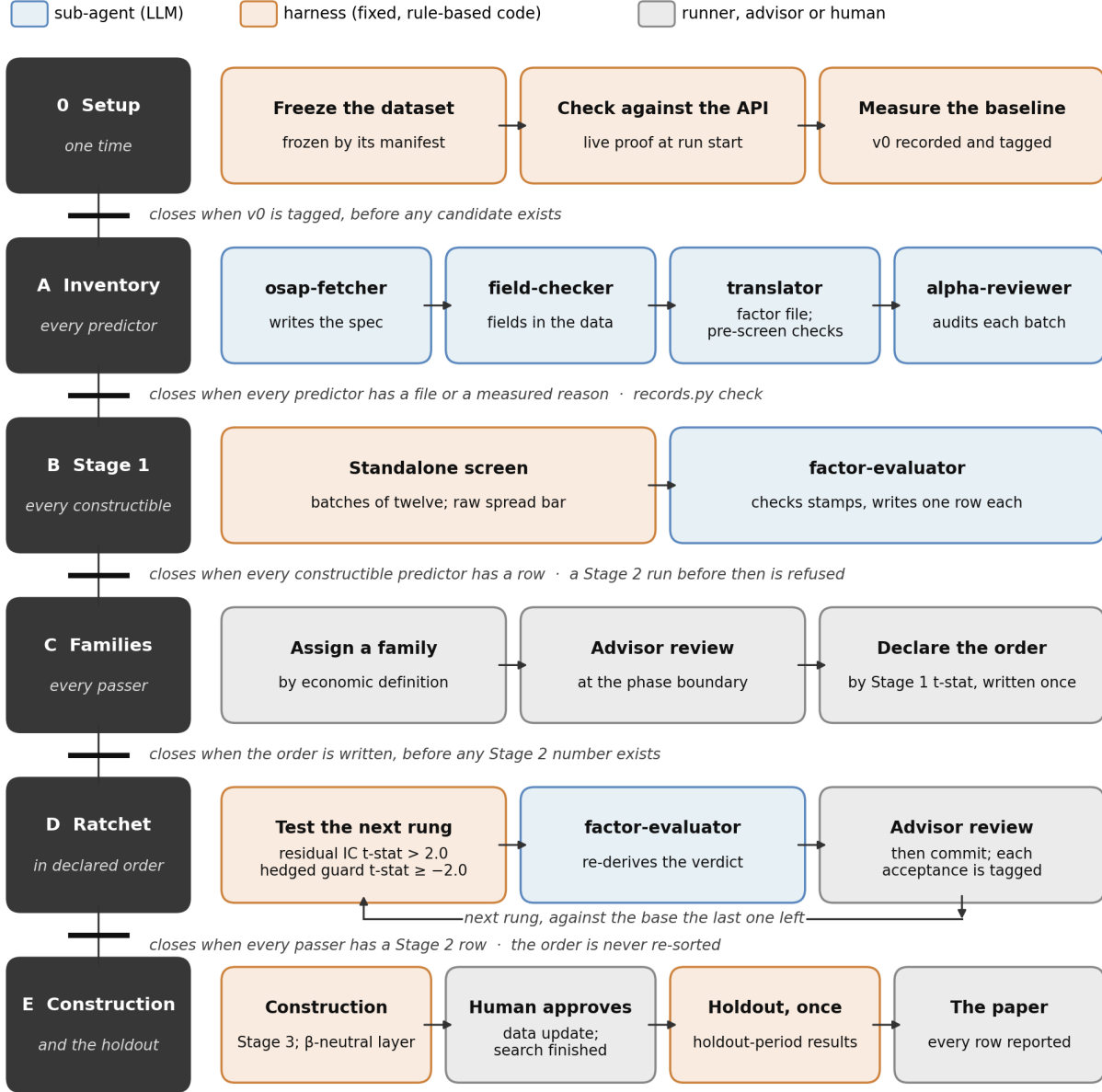


Figure 2. The phase machine. Each row is a phase, read left to right; the bar between two rows is the condition that closes the phase above, and no phase starts before it holds. Colour shows who does each step: a sub-agent, the fixed, rule-based harness, or the runner, advisor or human. In the ratchet, each rung is tested against the composite the previous rung left.

E. The Inner Cycle

Within a phase, the runner repeats the same cycle on every turn. It reads the current state and finishes any open item before starting a new one. It declares the next batch in the event log, with its members and the stamps it will run under. It runs the harness. When the run finishes, it

hands the result to the evaluator, which checks the four stamps against the repository, applies the bars, confirms that its own verdicts equal the ones the harness stamped, and writes one registry row per candidate along with the events that record the decision. Before committing a ladder's acceptances the runner consults the advisor, and the evaluator then implements each new version: it moves the factor file into the accepted set, edits the composite, runs a fresh baseline and a construction run for the new version, and appends an entry to the model manifest. The runner then commits the step with the four stamps in the commit message, adds one annotated tag per new version, and updates the state file. Only then does the runner reply, and the reply ends with the status summary described above. Figure 3 draws the cycle. A rejection takes the short path: the rows are written, the composite is unchanged and the step is committed.

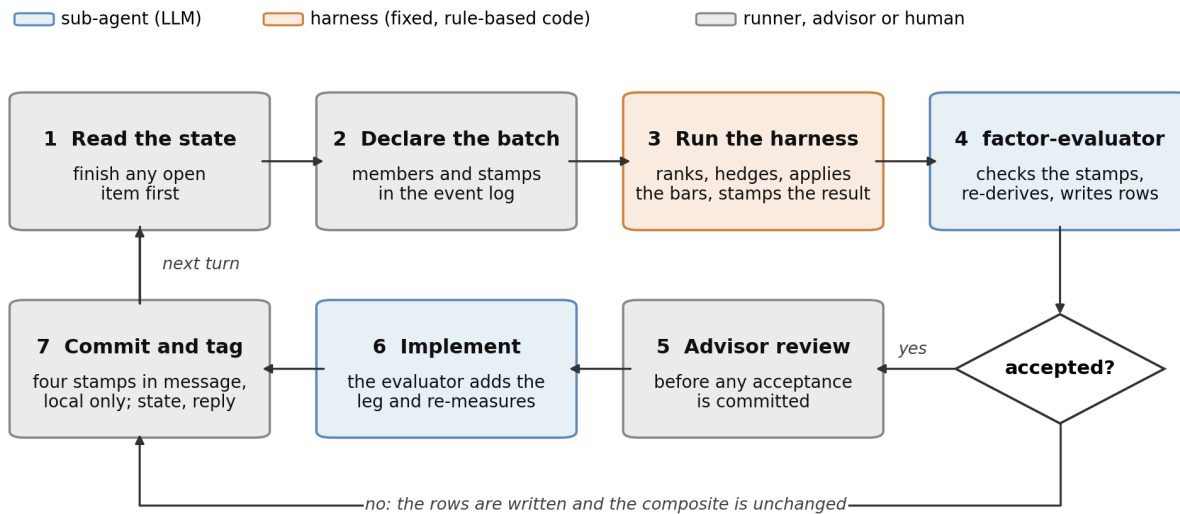


Figure 3. The inner cycle, one turn of the loop that measures something. Steps 1 to 4 run on every such turn. An acceptance adds the advisor's review and the implementation of a new version (steps 5 and 6); a rejection goes straight to the commit. Every turn ends committed locally, with the state file updated, so the next turn, or the next session, starts from the record.

The cycle is sequential on purpose. In the ratchet, rung $i+1$'s base depends on rung i 's verdict, and every accepted version must be measured and tagged before the next ladder is drawn up. Parallelism is used only where it cannot change a result: the inventory ran several fetches at a time, because a specification depends on nothing but its own reference code.

F. Example: One Predictor through the Loop

PctAcc, the percent accruals of Hafzalla, Lundholm, and Van Winkle (2011), went first through the ratchet, and following it shows how the parts fit together.

The fetcher wrote its specification from the reference code at the pinned commit. The construction is income before extraordinary items minus operating cash flow, scaled by the absolute value of income, with low accruals attractive. The catalogue's code has a balance-sheet fallback for firms without a cash-flow statement that needs income taxes payable, which Sharadar does not publish; the loop logged a decision not to reproduce that branch, at a measured cost of about 0.3% of the universe, rather than approximate it. The translator wrote the factor file, and it passed the pre-screen data checks.

PctAcc was screened in the seventh batch of twelve. It passed every bar, with a mean IC of 0.0112 and a Newey-West t-stat of 4.03, positive in both halves (0.0152 and 0.0071), a raw long-short of 4.74% a year and coverage of 95.6%. The evaluator also logged what the bars do not test: the IC level cleared its bar by a thin margin, the second half was under half the first, and the information sat in the mid and small tiers. It was then assigned to the investment family, because the decision that fixed the label-to-family map absorbed the catalogue's accruals label into investment. The order of the ratchet was declared at the same time, and PctAcc came first, because its Stage 1 t-stat was the highest of the 24 passers.

It was the first rung of the first ladder, tested against the five-leg baseline. Its residual IC t-stat was 4.67 against a bar of 2.0, and its guard t-stat was 0.09 against a floor of -2.0, so it was accepted. The composite's IC rose with a paired t-stat of 2.81 while its long-short barely moved, and the evaluator recorded why: the information showed in the ranking, not in the spread. The evaluator re-derived the verdict from the stamped results, and the runner consulted the advisor before committing. The version was then adopted as v1: the file moved to the accepted set, the composite gained a sixth leg, the fresh baseline reproduced the with-candidate model tested on the rung in every compared statistic to six decimal places, and the construction run was added to

the manifest. From specification to version took less than a day, and every step left a timestamped event, a file and a commit that a reader can check.

G. Provenance and Immutability

Four hashes go on every set of results: HARNESS_SHA over the harness code, CONFIG_SHA over the pre-registration file, COMPOSITE_SHA over the composite definition together with its accepted legs, and DATA_SHA over the manifest of the frozen dataset. The construction layer adds a fifth, LAYER_SHA, over its parameter file. The evaluator's first step on any run is to compare the result's stamps with the repository's. On a mismatch it logs the fact and writes nothing, so a number produced by other code, other rules or other data cannot enter the record. The harness also stamps its own verdict into each result, and the evaluator re-derives every verdict from the result's raw statistics. A disagreement is treated as a harness defect, not as a rounding difference.

Across all 56 completed runs, CONFIG_SHA and the universe hash each took exactly one value, so the pre-registration never moved between the first baseline and the holdout. DATA_SHA took two: the first download, on which every decision of the search was taken, and the data update that the holdout protocol required, which I approved as a stop-and-ask question. HARNESS_SHA moved 5 times: once at bootstrap before any run, twice for the construction layer and the fixes its review asked for, and twice for diagnostics added after the ratchet closed. None of the moves fell between the first screen and the close of the ratchet, and every harness that later produced a number was first shown to reproduce the live composite on every compared statistic (Appendix E).

Git hooks make the record hard to rewrite. The pre-commit hook refuses the vendor data files, anything that looks like a credential or an API key, any change to the harness, a factor or the configuration that does not pass the test suite, and a results file that is still being written. It also refuses an edit to a results file that has already been committed. The commit-message hook requires a type prefix and the four stamps. Each composite version gets one annotated tag, which

is never moved, and mistakes are undone with a new commit rather than by rewriting history. The permission layer asks the human before a hard reset.

H. Memory and Continuity

An LLM forgets everything between sessions, and a search of this size runs across many sessions. The loop therefore keeps its memory in files, and each file has one job. A state file of at most 25 lines holds the current state only: the phase, the live composite and its stamps, the counters, the open items and the next step. Its history goes to a separate file, so the state stays short enough to read in full at the start of every session. The event log is append-only and holds numbers and identifiers rather than prose. It had 1134 events when the search closed, and every decision in this paper can be traced to one of them. A registry holds one row per candidate with its bars and verdict, and an index gives one line per candidate. The model manifest holds one entry per composite version. A decisions file records every design decision with its date, owner and evidence, and a journal records each step with what it ran and what it cost.

The records checker ties these together. It rebuilds the index from the registry, reconciles the catalogue against candidate files, registry rows and logged exclusions, and prints the predictors still open. It flags an inconsistency when a phase checkpoint is violated, when a completed run is unevaluated, or when the proof that the dataset still matches the vendor’s API is missing, stale or taken on a different data hash. A fresh session starts by reading the state file and running the checker, and from there it can resume in the middle of a phase. Operational lessons that are not rules of the search are kept as short notes that later sessions load automatically.

I. The Autonomy Boundary

The runner stops for a human on seven questions only: running the out-of-sample test; anything that moves CONFIG_SHA; anything that moves DATA_SHA; moving the pinned reference-code commit; declaring the search finished; a rule contradicting itself, or a result implying that logged rows are wrong; and opening a tenth family. Everything else it decides, logs the reasoning, and carries out.

The permission layer backs the list with a second line of defence. Commands the loop needs are allowed without asking: running the harness, the pre-screen data checks and the record scripts, running the tests, writing under the research, factor, harness, documentation and source directories, and committing and tagging. Actions that correspond to the stop-and-ask list require the human's approval each time: editing the pre-registration file or the data manifest, downloading data, running the out-of-sample test, or resetting history. Some actions are denied outright. These are reading the file that holds the API key, deleting the data, and reading any directory outside the project, so that nothing but this search's own measurements can inform a verdict.

Once the loop started, my role was limited to two things. First, I supplied the data: a Sharadar licence and the API key that downloaded the data. Second, I answered the stop-and-ask questions. Five of the seven arose, and I resolved them in three decisions. Before any run, the loop found that its decisions file contradicted itself on which bars the hedge should reach (question 6); I resolved it with a change to the configuration (question 2), under which the Stage 1 spread bar reads the raw long-short and the Stage 2 guard the hedged one (ruling D11). Later I approved the data update that the holdout protocol required (question 3), and at the end I confirmed that the search was finished and authorised the out-of-sample evaluation (questions 5 and 1). I wrote the design rules and fixed them before the loop started, in a bootstrap session that set up the repository and measured the baseline. During the run I did not edit any record by hand.

J. The Advisor

The runner is built to keep moving, and an agent that keeps moving can rationalise its way past a problem. The advisor is the counterweight. It is a stronger model that sees the runner's whole transcript, every tool call and every result, and is asked what is wrong with the current plan. The instruction file requires a consultation at every phase boundary, before every acceptance is committed, before any stop-and-ask question is raised, and whenever a result looks

too good, such as a single-factor t-stat above about 6 or a residual share near 100%. Advice that changes a decision is logged.

The event log records 6 consultations explicitly: before committing each of the four ladders that produced acceptances, at the boundary between the ratchet and the construction phase, and on the verdict that the data update could not have moved any Stage 2 decision. The decisions logged beside them set working rules rather than reversing verdicts: acceptances are applied one version at a time in ladder order, each version's baseline must reproduce its rung to six decimal places or the sequence stops, and the construction phase reads the Stage 3 run of the final composite rather than a new one.

Two more model choices belong on the record. The runner was Claude Opus (version 5.5 in the journal), operating in the Claude Code desktop application, after a bootstrap session on Claude Fable 5.1; the advisor was Claude Fable. The fetcher, field checker and translator are defined on Claude Sonnet, and the reviewer and evaluator run on the runner's model. None of these choices affects a number. The harness is the same Python code whichever model invokes it, and the evaluator re-derives every verdict from the stamped results rather than from any model's reading of it.

K. Why the Rules Take This Form

Most rules in this paper answer a specific way an automated factor search can go wrong.

Stage 2 accepts on information and prints spanning as a diagnostic. A return-based spanning test selects on portfolio return, which contradicts the aim of selecting on information, and it can reject a candidate that raises the composite's own IC. No transaction cost is charged anywhere in the search: a flat cost charged on standalone decile portfolios would penalise high-turnover signals, although a leg in a blend contributes only a fraction of its standalone turnover, so turnover is reported instead. Phase A inventories the whole catalogue, because a candidate list built from whichever files happen to have been translated leaves predictors unexamined without anyone deciding to exclude them. The out-of-sample period is 57 months long, because a tail of a

few months has no power. The rule against moving a stamp while a run is unevaluated covers every stamp, because a run is stranded whichever stamp moves. And the composite is a family blend with a residual-information bar, because a paired composite-IC bar is underpowered for one leg added to a blend of about nine legs, and an equal-weight blend lets near-duplicates take whole votes.

The construction follows the same logic. A ratchet that accepts on information and guards only the raw long-short can accumulate a large market beta that none of its rules measures, and a construction layer that is neutral to sector and to dollars but not to beta leaves that exposure in the book. This search therefore ranks every signal within its sector, hedges every long-short with a trailing beta, prints the beta and a date-free ex-regime Sharpe on every result, and constrains the construction layer’s beta.

These choices reflect prior knowledge of how factor searches behave over the same sample period, including the years reserved here as the holdout. The holdout is therefore a clean test of the selection made inside this search but not an unbiased test of the design choices themselves, and the decisions file says so (D2). Every candidate was judged on this dataset alone, and every field mapping was verified on it before use. Section XI returns to the design’s dependence on the window.

III. Data

The loop’s results rest on three inputs: the vendor data, the catalogue of predictors the candidates come from, and the monthly universe drawn from the data. This section describes each, and Table III summarises the sample.

A. The Sharadar Dataset

The data are 13 Sharadar tables, downloaded for this study with their full history, including 45.4 million rows of daily prices. The dataset was then frozen: a manifest records its files, and the manifest’s hash, DATA_SHA, is stamped on every result. A live check at the start of every run proves that the frozen data still match the API’s schema and history. Fundamentals are

read from Sharadar’s as-reported trailing-twelve-month reporting basis, taking the most recent filing as of the signal date and allowing it to be at most fifteen months old; a factor that needs quarterly or annual statements declares the exception in its file, and the harness prints every exception on the run. Where a Compustat line has no Sharadar equivalent, the factor file declares the gap rather than estimating around it. The data are licensed from Sharadar through Nasdaq Data Link and may not be redistributed, so the repository holds the manifest and the code that reads the tables, not the tables themselves.

The data were downloaded twice. The first download was full-history, so it already held the months reserved as the holdout: the holdout was isolated by rule, not by absent data. Every result of every decision run has an evaluation end of 31 December 2021, and a result that extends into 2022 is flagged as an out-of-sample breach. The holdout period ends in September 2026, after that download was taken, so the protocol required a data update before the out-of-sample evaluation. The second download took the same tables again and added one external table, the three-month Treasury bill rate (TB3MS) from FRED, for a diagnostic described in Section IV. The update restated the in-sample results mildly: on the same harness, configuration and composite, 54 reported statistics differ, the composite IC moved by -0.000009 , and the hedged Sharpe moved from 0.983 to 0.996. The loop judged, against the thinnest residual margins, that no Stage 2 verdict sat close enough to its bar to move; the rungs were not re-run. Unless labelled otherwise, the in-sample figures in Sections VI to VIII are on the original data, on which every verdict was taken, and those in Section IX are on the updated data, on which the holdout was read.

Panel B of Table III lists the tables held, with their row counts and the dates they span. The universe and the returns rest on four of them: daily prices and volume (SEP), daily market capitalisation (DAILY), the ticker master (TICKERS), which supplies the exchange, share class and sector, and corporate actions (ACTIONS), which classify delistings. Fundamentals come from SF1. The remaining tables, which hold insider transactions, institutional holdings, filing events, index membership and price metrics, serve individual predictors.

B. The Predictor Catalogue

The candidates come from the Open Source Asset Pricing (OSAP) catalogue of Chen and Zimmermann (2022), which publishes construction code and documentation for the cross-sectional return predictors of the published literature. The loop reads it at one pinned commit, b4e911e6 of release v2.0.0, never at the moving main branch, and the commit is recorded with the other stamps in Appendix E. At that commit the catalogue lists 212 predictors. For each one the loop uses two things: the reference code, which defines the construction, and the documentation row, whose economic category is the reference for assigning a family (Section IV.D). The reference code is written for CRSP, Compustat, IBES and other WRDS sources. Each predictor is therefore translated to Sharadar fields, and one whose inputs Sharadar does not publish is excluded with its reason (Appendix C) rather than approximated. Section VI accounts for all 212.

C. Universe and Sample

The universe is US common stock on NYSE, NASDAQ and NYSE American with a price of at least one dollar, rebuilt every month. A name enters at a market capitalisation at or above the NYSE 20th percentile and a 20-day median dollar volume at or above the 20th percentile of the cap-screened names; a current member leaves only when it falls below the 15th percentiles. The band removes churn at the boundary. The membership rule starts from scratch at the first month of the data, so a month's universe depends on configuration, data and month only, never on the order in which anything was run. In-sample the composite's deciles hold about 196 names on average. Liquidity tiers (MEGA, MID and SMALL, by volume percentile within the universe) are reported as diagnostics and never used for selection.

Table III
The Sample

Panel A describes the investable universe month by month, built by the harness's own screens on the updated data (DATA 42587e08609a) and grouped by period (paper/manuscript/build_sample.py). Names per month are the mean, minimum and maximum over the period's months. Market capitalisation is in dollars: the median is the time-series mean of each month's cross-sectional median, and the total is the mean of each month's sum. "Listed" is the same month's cross-section after the absolute screens alone (exchange, domestic common stock, a price of at least one dollar, a trade within seven days, a market capitalisation and a dollar volume), before the size and liquidity screens.

The exchange share is of names; NASDAQ and NYSE American hold the rest. The holdout row is descriptive and carries no model statistic. Panel B lists every table in the updated dataset with its rows and the dates it spans; tables without a date column show none. TB3MS is the one external table, added by the D8 data update. ACTIONS also lists corporate actions scheduled for dates after the download, which fall outside the holdout.

Panel A. The Universe by Period

period	months	names per month	min	max	distinct firms	firm- months	median cap, \$bn	total cap, \$tn	% of listed names	% of listed cap	% NYSE
1999–2004	72	2,207	1,887	2,858	4,731	158,883	1.24	13.0	38.5	95.7	56.2
2005–2010	72	1,853	1,772	1,944	3,159	133,431	2.12	14.5	39.2	95.3	62.9
2011–2016	72	1,822	1,730	1,943	3,029	131,176	3.22	20.6	43.9	96.4	63.5
2017–2021	60	1,951	1,841	2,312	3,337	117,046	4.13	34.4	46.7	97.5	59.3
In-sample, 1999–2021	276	1,958	1,730	2,858	7,499	540,536	2.62	20.0	41.5	96.5	60.3
Holdout, 2022–2026:09	57	1,928	1,737	2,317	3,024	109,874	5.48	56.2	44.7	98.3	57.7

Panel B. The Tables Held

table	content	rows	first date	last date
ACTIONS	corporate actions, including delistings	714,029	1997-12-31	2026-10-05
DAILY	daily market cap and valuation ratios	39,855,506	1998-12-01	2026-10-01
DESCRIPTIONS	field dictionary	386	—	—
EVENTS	8-K event codes	2,532,960	1993-11-08	2026-10-01
METRICS	price metrics: betas, 52-week range, averages	30,939	1997-12-31	2026-10-01
SEP	daily prices and volume	45,393,854	1997-12-31	2026-10-01
SF1	fundamentals, as reported and restated	3,218,127	1990-06-06	2026-10-01
SF2	insider transactions	11,559,783	2008-01-02	2026-09-30
SF3	institutional holdings (13F)	81,210,476	2013-06-30	2026-06-30
SF3A	13F holdings aggregated by stock	670,115	—	—
SF3B	13F holdings aggregated by investor	306,512	—	—
SP500	S&P 500 membership changes	60,185	1957-03-04	2026-10-01
TB3MS	three-month Treasury bill rate (FRED), monthly	1,113	1934-01-01	2026-09-01
TICKERS	ticker master: exchange, share class, sector	74,282	—	—

Over the 276 in-sample months the universe holds 1,958 names in the average month, never fewer than 1,730 or more than 2,858, and 7,499 distinct firms over 540,536 firm-months. That count equals the forward returns the composite’s in-sample run measured, name for name. The size and liquidity screens remove more than half of the listed names but little of the market’s value: the universe holds 41.5% of the names that pass the absolute screens and 96.5% of their market capitalisation. NYSE names make up 60.3% of the universe and NASDAQ names 39.3%. The universe is largest in 1999 to 2004, at 2,207 names on average, and its median member

grows from \$1.24 billion in that period to \$4.13 billion in 2017 to 2021. The 57 holdout months hold 1,928 names on average.

IV. Methodology: What the Harness Measures

This section and the next describe the harness: the fixed measurement that every candidate faces and that no agent can change without the change showing on every later result. Two parts of it deserve particular attention, the within-sector ranks and the market hedge; both were fixed before any candidate was screened.

A. Signals, Ranks and Forward Returns

A signal is observed at the prior business month-end and held for the following month. Within each month it is winsorised at the 1st and 99th percentiles and converted to a percentile rank within the name's sector, oriented so that a high rank is attractive according to the catalogue's published sign. A sector-month with fewer than 10 scored names falls back to the cross-section rank. Ranking within sector means every signal compares a firm with its peers: a value signal does not become a bet on cheap industries, and the composite does not load on sectors by construction. The sector is Sharadar's current classification applied to every month, a look-ahead the decisions file states (D3). The forward return runs from the signal-date close to the next business month-end, using Sharadar's adjusted close. Delistings follow Shumway's (1997) convention by reason: a name that leaves for a performance reason (bankruptcy, a regulatory delisting) is assigned a return of -30% for its final partial month, and one that leaves through a merger or voluntary delisting earns its last observed price. Performance delistings are rare in this universe, 21 name-months out of 540,536 in the final in-sample run against 2,342 merger delistings, so the convention is rarely triggered.

B. The Information Coefficient and Its Newey-West t-Statistic

Every selection decision in this paper rests on one statistic, the monthly rank information coefficient, and on a Newey-West t-statistic of its mean, so it is worth being precise about both.

In month t the IC is the Spearman rank correlation, across all names with both a score and a forward return that month, between the signal's within-sector percentile rank and the forward return. It is computed on the continuous score, not on decile membership, so that within-decile ordering counts. A month needs at least ten scored names to produce an IC. Over the decision window this gives a series of up to 276 monthly values (fewer for a signal with a minimum-history requirement). The mean of that series is the mean IC reported everywhere in this paper; the ICIR is the mean divided by the standard deviation of the series.

Monthly IC series are autocorrelated, so the t -statistic of the mean uses the Newey and West (1987) long-run variance with a Bartlett kernel. With demeaned values e_t , T months and L lags, the long-run variance is

$$\text{LRV} = \frac{1}{T} \left[\sum_{t=1}^T e_t^2 + 2 \sum_{l=1}^L \left(1 - \frac{l}{L+1} \right) \sum_{t=l+1}^T e_t e_{t-l} \right],$$

and the t -statistic is the mean IC divided by the square root of LRV/T . The lag is registered in the configuration as $L = 3$ and is the same for every t -statistic in the search: the IC series, the long-short return series, and every paired difference. It was fixed before the first screen.

The Stage 2 residual IC is built the same way on a transformed score. In each month the candidate's percentile rank is mapped to a normal score, and so is each current leg's; the candidate's normal score is regressed, with an intercept, on the legs' normal scores (a name that lacks a leg is set to that leg's cross-sectional mean of zero); and the residual's Spearman correlation with the forward return is that month's residual IC. The series' mean and Newey-West t -stat follow as above. The residual IC measures the information the candidate carries that no linear combination of the existing legs carries, and it does not depend on how the legs are weighted. The residual share reported on every rung is the mean residual IC divided by the candidate's own mean IC.

C. The Long-Short and the Market Hedge

The raw long-short return of a signal is the equal-weight return of its top decile minus its bottom decile, formed each month on the same ranks, gross of any cost. A decile spread is not market-neutral. Signals that favour low-volatility, illiquid or profitable firms hold a long side with less market exposure than the short side, so their raw spread is short the market, and a ratchet that accepts on information can accumulate that exposure leg by leg without any bar seeing it. The harness therefore reports every long-short hedged to the market (D4):

$$LS_t = (R_{D10,t} - R_{D1,t}) - \beta_t M_t,$$

where M_t is the universe's own capitalisation-weighted total return and β_t is the slope of the raw spread on M estimated over months $t-36$ to $t-1$ only, set to zero before 12 months of history exist. The beta is ex ante: no month's hedge uses that month's return. Both series are annualised as the monthly mean times twelve, so the difference between them is exactly -12 times the mean of $\beta_t M_t$. This paper calls that difference the hedge term. For a book that is short the market ($\beta_t < 0$) in a rising market the hedge term is positive: the hedge adds a long market position to a net-short spread.

The hedge reaches the bars in exactly one place. The Stage 1 positive-spread bar reads the raw spread, and the Stage 2 guard reads the hedged family blend (ruling D11, Section V). Every headline in the paper is hedged, and every result prints beside it the raw series, the ex-ante beta averaged over the window, the full-window beta of the raw spread, and two diagnostics that never decide (D5). The first is a date-free ex-regime Sharpe: the hedged Sharpe after removing the 3 best calendar years of the long-short, whichever they are. The second splits the months by the sign of the market's trailing 12-month return into a bear and a bull Sharpe.

Two properties of the hedge were known, disclosed and left in place, because changing the hedge would move the pre-registration. First, M is a total return, so a negative-beta book is credited $|\beta_t|$ times the risk-free rate as if it were return. After the data update added the Treasury bill rate, the harness reports beside every hedged series an excess-of-rf diagnostic, $LS_t + \beta_t \cdot rf_t$ with rf the three-month bill rate of the holding month, and the difference, the rf credit, in

percentage points a year; it is a diagnostic and decides nothing. Second, the guard inherits the hedge term: a candidate that pushes the blend's ex-ante beta down is credited through the hedge whatever its information, so for a negative-beta leg the guard is close to non-binding and the residual IC is the bar that decides (decision hedge_guard_negative_beta_property, logged during the ratchet).

The paired change in long-short return that the Stage 2 guard reads is the month-by-month difference between the hedged family blend's long-short return with the candidate and without it, and its t-statistic is the Newey-West t-stat of that difference series. Sharpe ratios are annualised from monthly values; long-short t-statistics are Newey-West as above.

D. The Family Blend

A flat equal-weight blend lets near-duplicates take whole votes: three low-volatility signals would carry three times the weight of one. The composite here is a two-level blend. Every Stage 1 passer is assigned, by its economic definition, to one of fewer than ten families, and the composite score is the equal-weight mean of family scores, each family score being the equal-weight mean of its members' within-sector percentile ranks. Both means are taken over the legs a name actually has, so a missing leg neither drops the name nor pulls it to the middle. A new leg therefore dilutes only its own family. Families are assigned after a candidate's screen and before any Stage 2 number exists, with a timestamp, and are never changed afterwards. The five seed families are fixed by the baseline: size, value, profitability, investment and momentum. The reference for assignment is the catalogue's own economic category, read against the construction.

Table IV
The Nine Factor Families

The five seed families are fixed by the baseline; the four others were opened by Stage 1 passers during the family phase. Members assigned counts the seed leg and every Stage 1 passer assigned to the family (research/families.yaml). A family's weight in the composite is one over the number of families with a live leg; within a family every leg has equal weight. The accruals and volume labels were absorbed into investment and liquidity by definition (decision phase_c_family_partition), which kept the count at the cap of nine.

family	origin	definition	members assigned	live legs in v14	weight in v14
size	baseline	market capitalisation (small = attractive)	1	Size	0.1111
value	baseline	price relative to a fundamental anchor	4	Value, cfp	0.1111

family	origin	definition	members assigned	live legs in v14	weight in v14
profitability	baseline	earning power relative to capital	6	Profitability, CBOperProf, GP, roaq, RoE	0.1111
investment	baseline	growth of the asset base or of investment	2	Investment, PctAcc	0.1111
momentum	baseline	continuation of past returns	2	Momentum, TrendFactor	0.1111
external financing	opened in Phase C	net capital raised from or returned to investors (share issuance, net equity and debt financing)	4	ShareIss5Y, XFIN	0.1111
volatility	opened in Phase C	dispersion or tail size of a stock's own returns (total, idiosyncratic, extreme daily); low = attractive	4	MaxRet, IdioVol3F	0.1111
short-term reversal	opened in Phase C	reversal of the most recent month's return	1	STreversal	0.1111
liquidity	opened in Phase C	trading activity and trading cost (turnover, zero-volume days, bid-ask spread, volume trend)	5	zerotrade6M, VolumeTrend	0.1111

The baseline, v0, is the four stock-level characteristics behind the Fama-French five-factor model (the market factor has no cross-sectional counterpart) plus twelve-month momentum, as rank signals with one family each: small size, high book-to-market, high operating profitability, low asset growth, and past winners with the most recent month skipped. It was measured and tagged before any candidate was screened.

V. The Selection Rules

Every parameter in Table V is fixed in a single configuration file and hashed into CONFIG_SHA. Changing any of them changes the pre-registered specification, which only the human can authorise.

Table V
The Fixed Harness

The measurement settings every candidate is tested under, from config/test_config.yaml (CONFIG_SHA 0d88328d5b10). Changing any row changes the pre-registered specification, which is a human decision.

Element	Specification
Data	Thirteen Sharadar tables, full history, frozen by a manifest (DATA_SHA). A live check at every run start proves the frozen data still match the vendor's live database. Updated once, under D8, before the out-of-sample evaluation; the update added the three-month Treasury bill rate (TB3MS) for a diagnostic.
Universe	US common stock on NYSE, NASDAQ and NYSE American, price $\geq$ \$1. Entry at the NYSE 20th percentile of market cap and the 20th percentile of dollar volume; exit below the 15th percentiles. Rebuilt monthly; the membership rule starts from scratch at the first month of the data.
Decision window	January 1999 to December 2021, 276 monthly rebalances, fixed and absolute.
Out-of-sample period	January 2022 to September 2026, 57 months. Reserved; a result extending into it is an out-of-sample breach unless it is the final validation; evaluated once on the finished composite.
Rebalance	Monthly. Signal observed at the prior business month-end and held for the following month.

Element	Specification
Point-in-time fundamentals	Sharadar's as-reported trailing-twelve-month reporting basis (ART), most recent filing as of the signal date, at most 15 months old; a factor that needs quarterly or annual data declares the exception in its file.
Returns	Sharadar's adjusted close; Shumway delisting convention by reason (−30% on performance delistings, last price otherwise).
Signal construction	Winsorised at the 1st and 99th percentiles, then percentile-ranked within the name's sector each month; a sector-month with fewer than 10 scored names falls back to the cross-section rank (D3). Composite: two-level family blend, equal weight across families and within, renormalised over the legs a name has.
Market hedge	Every long-short is reported hedged: $D10 - D1 - \beta_t \times M_t$, where M is the universe's own cap-weighted total return and β_t is estimated on months $t-36..t-1$ only (0 before 12 months) (D4). The raw series, the ex-ante and full-window β , and the Sharpe ex the three best calendar years print beside it (D5).
Statistics	Monthly Spearman IC of the continuous score against next-month return; every t-statistic Newey-West with 3 lags. Equal-weight decile 10 minus decile 1 long-short with annualised Sharpe, drawdown, hit rate and turnover; coverage; halves, annual, tier and decay diagnostics. No transaction cost is charged anywhere in the search.
Minimum sample	Fewer than 120 usable long-short months is inconclusive rather than rejected; at least 30 names per decile.

Stage 1, the Standalone Screen

Stage 1 is a screen. Its only job is to stop weak or broken factors from consuming a ratchet run, and passing it earns a family assignment and a Stage 2 rung, nothing more. A candidate passes when all of the following hold on the decision window: a mean monthly IC of at least 0.010; a Newey-West IC t-statistic of at least 2.50; a positive mean IC in each half of the window; a positive gross annual return on the raw decile-10-minus-decile-1 spread; coverage of at least 40% of universe name-months; and at least 30 names per decile on average. The spread bar reads the raw series on purpose: a hedged spread bar would pass a signal whose own spread loses money but whose short-market tilt earns the hedge term. A candidate whose published sign fails but whose reversed sign passes is treated as a second hypothesis: it must clear an absolute t-stat of 2.74 together with every other bar, as a separately declared file.

Stage 2, the Ratchet

Stage 2 asks whether the candidate adds information to the model that already exists. Two bars must both hold. The residual IC bar requires the Newey-West t-stat of the candidate's residual IC, as defined in Section IV.B, to be strictly greater than 2.0. The guard requires the Newey-West t-stat of the paired change in the hedged family blend's gross long-short return, with

the candidate against without, to be at least -2.0 , so that a signal which adds information but drags the blend's market-neutral return significantly is caught. Everything else printed on a rung, the paired composite change in IC, the spanning alpha of the candidate's long-short on the composite's, its R^2 , the change in Sharpe and drawdown, the raw and hedged changes in return and the change in beta, is a diagnostic. Diagnostics are reported on every rung and never decide.

The test is deliberately an information test rather than a return test. One leg added to a many-family blend moves the blend by little, so a paired test of such a change is underpowered, and a return-based spanning bar can reject candidates that raise the composite's own IC significantly (Section II.K).

Table VI
Acceptance Bars

The bars were fixed before the first candidate was run and are unchanged since. Both Stage 2 bars must hold. The Stage 1 spread bar reads the raw long-short; the Stage 2 guard reads the hedged family blend (D11).

Stage 1, standalone screen	Bar	Stage 2, marginal information	Bar
Mean monthly rank IC	≥ 0.010	Residual IC, NW t-stat	> 2.0 , strictly
Newey-West IC t-statistic	≥ 2.50	Guard: paired Δ LS return of the hedged family blend, NW t-stat	≥ -2.0
Mean IC in each half of the window	> 0	Paired Δ IC, spanning alpha, R^2 , Δ Sharpe, Δ MaxDD, β and ex-regime rows	diagnostics only
Raw gross D10–D1 annual return	> 0		
Coverage of universe name-months	$\geq 40\%$		
Average names per decile	≥ 30		
Reversed-sign screen (second hypothesis)	absolute t-stat ≥ 2.74 and every other bar		

Order, Ladders and Re-Derivation

Candidates are screened twelve at a time in alphabetical order of their catalogue acronym, never by any number. Every Stage 1 passer gets a Stage 2 rung. The order of the rungs is descending Stage 1 t-statistic over all passers, declared once when the family phase closes and written to a file before any Stage 2 number exists; the strongest signal in a family is thereby its representative, and weaker variants must add information beyond it. Ladders of at most five rungs are drawn from that list in order, and rung i faces the composite as it stands, the base plus

every earlier rung that passed. Two rules are enforced by code: the order is never re-sorted on a Stage 2 result, and HARNESS_SHA never moves inside a declared ladder.

Because rung $i+1$'s base depends on rung i 's verdict, the harness applies both bars itself and stamps its decision into each result. That is not a transfer of the decision. The evaluator re-derives every verdict from the result's raw statistics, and a disagreement is treated as a harness defect rather than a rounding difference. Across all 24 rungs the evaluator re-derived every verdict and none was overturned. Each accepted version was then implemented one at a time: its file moved, the composite edited, a fresh baseline and a Stage 3 construction run, a manifest entry appended, one annotated git tag. Every version's own baseline equals the with-candidate model tested on the rung that accepted it, on 66/66 compared statistics to six decimal places (Appendix E).

Rules Committed in Advance, and Rulings Made inside the Loop

The rules in Table VII were settled before they were needed, so that applying one is not a fresh choice made while looking at a result one would like to accept. The one ruling that changed how a bar is read, D11, was made at bootstrap before any run, as a stop-and-ask question that I answered. The others were made by the runner inside the loop as cases arose and are logged as decision events: how a family partition maps catalogue labels to families, when a qualified reversal need not be screened, how the guard's hedge bias is disclosed, and which bar a multi-bar failure is attributed to. Each was logged after the case that raised it and none introduced a new rule. I say so plainly because the honest claim is that the bars predate the screen and the readings of them do not.

Table VII
Rules Committed in Advance and Rulings Made inside the Loop

Each rule fixed before the search and each ruling made during it, with the incident behind it. D-numbers refer to the decisions file in the repository; other ids are decision events in research/events.jsonl.

Rule	Why it exists	Where it was triggered
Inventory everything first; exclude only for a measured reason	A candidate list built from whichever files exist silently leaves predictors unexamined, and a search that proposes its own candidates is selecting before it screens.	Phase A closed with every one of the 212 predictors accounted for: 5 seed legs, 106 screened, 101 excluded with a logged, measured reason (Section VI, Appendix C).

Rule	Why it exists	Where it was triggered
The hedge reaches one bar only (D11)	A hedged Stage 1 spread bar and a raw-only guard were both on the table; the decisions file contradicted itself.	Found at bootstrap, before any run, and put to me as stop-and-ask 6: Stage 1 reads the raw spread, the Stage 2 guard the hedged blend; the resolution moved the configuration (question 2), before any run.
A reversed sign is a second hypothesis	Flipping a sign after seeing the number doubles the hypotheses. A flipped screen faces an absolute t-stat ≥ 2.74 and every other bar, as a separately declared file.	BidAskSpread failed at a t-stat of -2.77 ; its reversal passed at 2.76 and was rejected at Stage 2. GrLTNOA qualified at an absolute t-stat of 2.75 but its reversed IC, 0.0059 , could not clear the IC bar, so it was not screened (flip_not_screened_when_deterministic_fail).
Families are assigned by definition, before any Stage 2 number, and never changed	A family fitted to the data after the ratchet is a selection.	One decision fixed the label-to-family map before any assignment (phase_c_family_partition); Stage 1 numbers were visible when it was chosen, and that is disclosed.
Registered bars decide and diagnostics report	A diagnostic disagreeing with a bar is not a rule contradicting itself; a new guard would move CONFIG_SHA, which is a human decision.	The guard's bias toward negative- β legs was disclosed rather than repaired (hedge_guard_negative_beta_property); XFIN was accepted with every diagnostic adverse (Section VI).
Inconclusive is a verdict distinct from rejected	A thin sample, a decile collapse or coverage under the floor says nothing about the factor.	Never triggered in-sample. The holdout-only cross-check drew the 120-month floor's warning on a 57-month period and no verdict.
Fix a harness defect before the next decision depends on it; never move a stamp while a completed run is unevaluated	A defect that produces a plausible wrong number is inherited by every later comparison.	HARNCESS_SHA moved 5 times, never between the first screen and the close of the ratchet, and every harness that later produced a number first reproduced the live composite (Section II.G, Appendix E).
The holdout is read against expectations written before the out-of-sample evaluation	A reading chosen after the number is data snooping.	D8 declared the out-of-sample protocol before the first screen; the event log records the expectations before the first holdout run (Section IX).
Every turn ends with a run and a decision, never a question	The only questions the loop may ask are the seven stop conditions.	Standing rule; five of the seven arose, and I resolved them in three decisions (1, 2, 3, 5 and 6).

VI. The First Complete Run

This section follows the loop through its first complete run, phase by phase. It is a case study of the framework: what each phase did, what the agents found, and where the rules bound.

Phase A: The Inventory

The Chen and Zimmermann (2022) catalogue lists 212 predictors at the pinned commit, and every one is accounted for. Five are the baseline legs. Of the rest, 106 were fetched, specified, field-checked, translated, run through the pre-screen data checks and screened; there were 107 screens, because BidAskSpread's declared reversal was screened as a separate file. The remaining 101 were not tested, each for a measured or structural reason. Of these, 64 need inputs Sharadar does not publish: IBES analyst data in 20 cases, options data in 10, and otherwise gross PP&E, advertising expense, income taxes payable, 13F holdings before 2013, governance

indices, credit ratings, business segments, short interest and pension footnotes. Another 28 failed the pre-screen data checks on this dataset, on coverage, on a mass point of tied values, or because they are binary indicators or discrete scores that cannot fill ten deciles. The last 9 start too late for the 120-month minimum. The three components sum to 101, and with the five seeds and the 106 screened files they account for all 212 (Figure 5 and Appendix C).

The agents wrote 207 construction specifications from the cached catalogue source, and the adversarial reviewer audited every translated batch before it was screened: 20 reviews in the inventory phase and 23 across the search. These audits changed the factor files before any number existed. They found the one critical issue of the inventory, a return-window signal that had declared itself exempt from the minimum-history requirement, and a dozen major ones, among them a book-to-market variant that paired book equity and market equity up to seventeen months apart, an enterprise-value floor whose logic was inverted, and a debt filter missing from five signals that read debt. Each was fixed before the screen. The translator, for its part, found a sign convention in Sharadar's discontinued-operations field and recorded it as a known trap in the field map. The market series that three predictors need (the Pástor and Stambaugh (2003) liquidity innovation, the Kelly and Jiang (2014) tail factor, and the trend factor's moving-average coefficients) were rebuilt inside the harness from the dataset, on past data only.

Phase B: The Standalone Screen

The 107 screens ran in nine alphabetical batches of up to twelve on one harness hash. Of these, 24 passed, 83 were rejected and none was inconclusive. Of the 83 rejections, 79 were decided by the t-statistic bar and four by the IC level; 29 carry a negative t-stat, meaning the published sign points the wrong way on this universe and window; 9 carry a positive t-stat between 2.0 and 2.5 and missed narrowly, SP closest at 2.497. The four rejected on the IC level all cleared the t-stat bar: ChNNCOA, OPLeverage, RevenueSurprise and EarningsConsistency ranked the cross-section consistently but by less than a hundredth of a correlation within sectors. Many well-known signals fail here. Beta, six-month momentum and the seasonal momentum

variants, Accruals, earnings surprise, the 52-week high, Amihud (2002) illiquidity, and value ratios such as assets-to-market, the December book-to-market variant (BMdec) and earnings-to-price all fail at Stage 1 (Appendix B).

Three patterns in the passers set up the rest of the paper. The first is market exposure. 22 of the 24 passers have a negative full-window beta on their raw spread, and the volatility and illiquidity passers sit near -1 : IdioVol3F at -0.98 , RealizedVol at -1.16 , BidAskSpreadFlip at -0.98 , MaxRet at -0.86 . For those signals most of the hedged return is hedge term; the Stage 1 bar reads the raw spread, so none of them passed on it. Only STreversal (0.44) and TrendFactor (0.05) have a positive beta. The second is that near-duplicates cluster. Volatility, profitability and external financing each appear four or five times, and the three zero-trade variants are one low-turnover sort in three windows, with annual IC series that correlate at 0.99. The third is that a strong IC can go with a weak portfolio. The volatility passers rank the cross-section well but their raw long-short barely earns: IdioVol3F has an IC t-stat of 3.15 and a raw Sharpe of 0.138, BidAskSpreadFlip 2.76 and 0.046.

The reversed-sign rule was used once. BidAskSpread's published sign failed with a t-stat of -2.77 . Its reversal, at 2.76, cleared the raised 2.74 bar, was declared as a separate file named BidAskSpreadFlip, and passed every bar as a second hypothesis. GrLTNOA's reversal also qualified, at an absolute t-stat of 2.75, but its reversed IC, 0.0059, sits under the 0.010 level by a margin the rank reversal cannot close, so the loop logged that the flip would fail deterministically and did not screen it: a flipped screen faces every Stage 1 bar, not the t-stat bar alone.

Phase C: Families

The 24 passers and the five seeds carry 11 distinct catalogue labels against a cap of nine families. One decision fixed the label-to-family map before any assignment. Two labels were absorbed where an existing definition covers the construction: volume into liquidity, and accruals into investment. Four new families were opened: volatility, external financing, short-term reversal and liquidity. That makes nine declared families, the maximum, and size gained no

passer. The decision names the alternatives it rejected, and it records that every Stage 1 number was visible when the partition was chosen. All 24 assignments carry the same timestamp, which precedes the first Stage 2 run. The order of the ratchet was declared at the same moment and never re-sorted. Every family, its definition, its members and their eventual verdicts are in Appendix A (Table AII).

Phase D: The Ratchet

Five ladders tested all 24 rungs in declared order: 14 were accepted, 10 rejected, none inconclusive. Every rejection failed the residual bar, and no rung failed the guard: the lowest guard t-stat of the search was -1.26 , on RoE, which was accepted. Table VIII and Figure 4 show the arc.

Table VIII
Composite Versions

Statistics for 1999 to 2021, 276 months, gross of costs, equal-weight decile 10 minus decile 1 of the family blend, on the original data (runs 001–042, DATA 198b281de1a0, the data every verdict was taken on). Each version’s statistics are its own baseline run, which reproduces the with-candidate model tested on the rung that accepted its last leg in every compared statistic. The long-short is hedged unless labelled raw; β is the full-window β of the raw long-short on the universe’s cap-weighted return; ex-top-3 is the hedged Sharpe without the three best calendar years.

version	added	legs	fami- lies	mean IC	IC t-stat (NW)	IC half 1	IC half 2	LS Sharpe	ann. return %	MaxDD %	β (raw)	raw Sharpe	Sharpe ex top- 3	D10 turnover %/mo
v0	baseline	5	5	0.0145	2.62	0.0275	0.0016	0.600	7.32	-45.6	-0.14	0.719	0.159	28.0
v1	PctAcc	6	5	0.0169	3.02	0.0304	0.0033	0.614	7.38	-41.3	-0.12	0.737	0.202	27.6
v2	CBOperProf	7	5	0.0169	3.02	0.0312	0.0025	0.661	7.82	-37.1	-0.12	0.777	0.249	28.0
v3	ShareIss5Y	8	6	0.0215	3.98	0.0336	0.0095	0.848	10.29	-41.9	-0.19	0.874	0.465	26.3
v4	cfp	9	6	0.0249	4.51	0.0375	0.0124	0.996	12.02	-40.6	-0.26	0.913	0.614	25.2
v5	XFIN	10	6	0.0245	4.27	0.0380	0.0110	0.914	11.56	-47.8	-0.34	0.808	0.479	25.0
v6	GP	11	6	0.0242	4.23	0.0377	0.0107	0.925	11.72	-49.4	-0.32	0.843	0.490	25.2
v7	MaxRet	12	7	0.0305	4.49	0.0444	0.0166	0.951	13.95	-41.7	-0.65	0.622	0.623	43.4
v8	roaq	13	7	0.0320	4.50	0.0460	0.0180	0.967	15.18	-43.4	-0.72	0.613	0.604	44.1
v9	RoE	14	7	0.0323	4.50	0.0462	0.0183	0.932	14.87	-43.7	-0.73	0.579	0.581	44.3
v10	IdioVol3F	15	7	0.0326	4.48	0.0457	0.0195	0.919	15.25	-45.8	-0.75	0.578	0.575	39.8
v11	STreversal	16	8	0.0355	5.36	0.0475	0.0236	0.942	14.61	-39.5	-0.51	0.737	0.565	59.1
v12	zerotrade6M	17	9	0.0373	5.15	0.0467	0.0279	0.985	16.16	-41.1	-0.64	0.708	0.646	56.8
v13	VolumeTrend	18	9	0.0372	5.24	0.0467	0.0277	0.979	16.13	-41.6	-0.62	0.719	0.635	56.6
v14	TrendFactor	19	9	0.0389	5.68	0.0498	0.0281	0.983	16.12	-43.2	-0.56	0.773	0.650	58.1

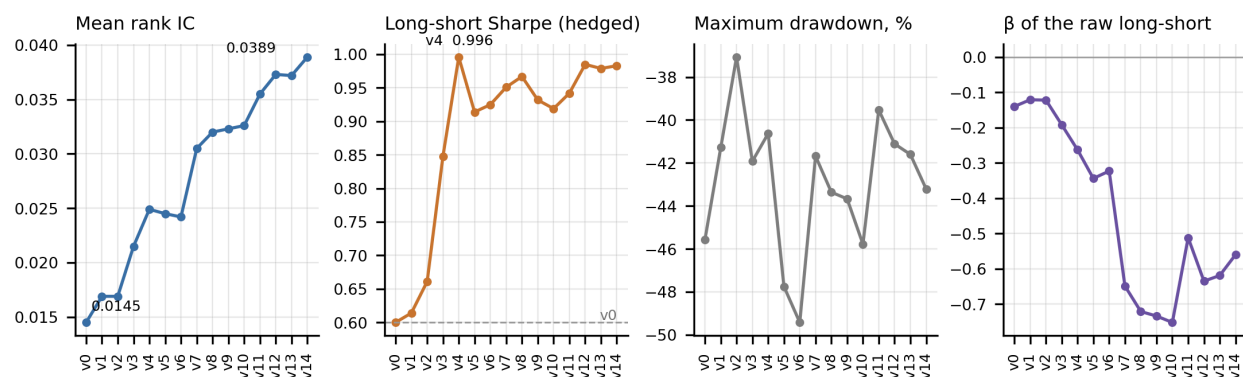


Figure 4. The ratchet by version. Mean IC rises at 10 of the 14 acceptances, as the residual bar makes likely. The hedged long-short Sharpe peaks at v4 and stays below that peak through the later ladders. The full-window β of the raw long-short falls from -0.14 to -0.75 by v10 and recovers to -0.56 as the two positive- β legs join; the hedge removes that exposure ex ante.

The first ladder accepted all five of its rungs. PctAcc, CBOperProf, ShareIss5Y, cfp and XFIN cleared the residual bar at t-statistics between 2.97 and 4.67, and took the composite from v0 to v5: the IC from 0.0145 to 0.0245, the hedged Sharpe from 0.600 to 0.914 through a peak of 0.996 at v4, and the second-half IC from 0.0016 to 0.0110. The beta of the raw spread went from -0.14 to -0.34 . The manifest carries a caveat on each of the five, and two deserve a sentence. cfp's portfolio spanning alpha on the composite had a t-stat of only 0.40 while its stock-level residual t-stat was 2.97; the two tests disagree and both are reported. XFIN cleared both bars while every diagnostic moved the wrong way: the paired change in IC had a t-stat of -0.62 , the hedged Sharpe changed by -0.082 , and the drawdown deepened. It was accepted on the registered bars.

The second ladder accepted GP, MaxRet, roaq and RoE and rejected ShareIss1Y, whose residual t-stat of 1.88 fell short because ShareIss5Y and XFIN already held its information. MaxRet is the clearest case of the hedge reaching the guard. It opened the volatility family at a full family weight. Its paired change in the blend's raw long-short was -0.59 percentage points a year, but the hedge part of the change was 2.82, because MaxRet pushed the blend's ex-ante beta down, so the guard read a hedged change of 2.23. The beta of the raw spread fell from -0.32 to -0.65 at v7, the raw Sharpe from 0.843 to 0.622, and top-decile turnover rose from 25.2 to 43.4% a month. Its residual t-stat of 4.01 carried the acceptance on its own, and the loop logged the

guard's bias as a disclosed property rather than inventing a raw-return guard, which would have moved the pre-registration. RoE was the ladder's weakest acceptance: every long-short diagnostic worsened, its guard t-stat of -1.26 was the lowest of the search, and its residual share was a quarter of its own IC. By v9 the composite IC was 0.0323, the raw beta -0.73 , and the raw Sharpe 0.579.

The third ladder accepted IdioVol3F and STreversal and rejected OperProfRD, NetEquityFinance and CF, whose information the profitability, financing and value legs already held (residual t-stats -0.42 , 1.40 and 1.02). IdioVol3F was the thinnest pass of the search, at 2.03, against a base that already held MaxRet. STreversal, the one-month reversal of Jegadeesh (1990), was the opposite case to MaxRet. Its residual IC exceeded its own IC (a residual share of 1.06), which triggered the reviewer's rule for results that look too good; the audit found the projection was removing a leg component that worked against the reversal, and cleared it. As a positive-beta leg it unwound part of the hedge: the raw change in the blend's return was $+1.09$ percentage points a year and the hedged change -0.64 , the beta of the raw spread rose from -0.75 to -0.51 , the IC t-stat rose from 4.48 to 5.36, and top-decile turnover went from 39.8 to 59.1% a month. No cost is charged in the search, so turnover is reported and never decides.

The fourth ladder accepted zerotrade6M, VolumeTrend and TrendFactor and rejected zerotrade12M and RealizedVol. zerotrade6M opened the liquidity family, the ninth and last; a tenth would have needed my approval. Its hedged Sharpe gain, from 0.942 to 0.985, was carried by the hedge: the raw Sharpe fell from 0.737 to 0.708 and the beta of the raw spread went to -0.64 . VolumeTrend passed at 2.04, the second-thinnest margin, and TrendFactor, the moving-average forecast of Han, Zhou, and Zhu (2016), at 2.14; both score only from January 2003, because the price history begins in December 1997 and their look-back windows are long. The fifth ladder, four rungs, accepted nothing: BidAskSpreadFlip (1.77), IdioVolAHT (1.92, the nearest miss of the search), zerotrade1M (-1.28) and NetPayoutYield (0.12) all failed the residual bar.

The declared order decided between near-twins, as it was designed to. Within the zero-trade cluster, zerotrade6M had the highest Stage 1 t-stat and was tested first; zerotrade12M then failed at 0.49 against a base holding it, and zerotrade1M at -1.28 . Within volatility, MaxRet came first and was accepted, IdioVol3F cleared narrowly, and RealizedVol and IdioVolAHT failed against a base holding both. A reader who doubts the order can see in Table AI what each rejected twin carried beyond the legs it faced.

The margins are worth stating once. The two thinnest acceptances were IdioVol3F at 2.03 and VolumeTrend at 2.04; the two nearest misses were IdioVolAHT at 1.92 and ShareIss1Y at 1.88. For scale, the data update moved the composite IC t-stat by -0.0019 . In every case the rule decided, and no rung was re-litigated.

v14 is the final composite: 19 legs in 9 families. On the original data its IC was 0.0389 with a t-stat of 5.68, its hedged Sharpe 0.983, its raw Sharpe 0.773 and the beta of its raw spread -0.560 . The hedged Sharpe peaked at v4; the ten acceptances after it raised the IC from 0.0249 to 0.0389 and its t-stat from 4.51 to 5.68, while the hedged Sharpe stayed between 0.914 and 0.985. On the updated data the composite's three best years, 2000, 2001 and 2021, carry 53.6% of the summed long-short, and its hedged Sharpe without them is 0.666. That is the whole-model statement of the regime concentration the rows had carried, and it was recorded before the holdout was evaluated.

The Rejection Table

Appendix B lists all 83 Stage 1 rejections with their IC, t-stat, halves, raw long-short Sharpe and return, beta, coverage and decisive bar. Table AI lists all 24 ratchet rungs with both bars and every diagnostic. Appendix C lists the 101 predictors not tested and the reason. Together they account for every predictor in the catalogue.

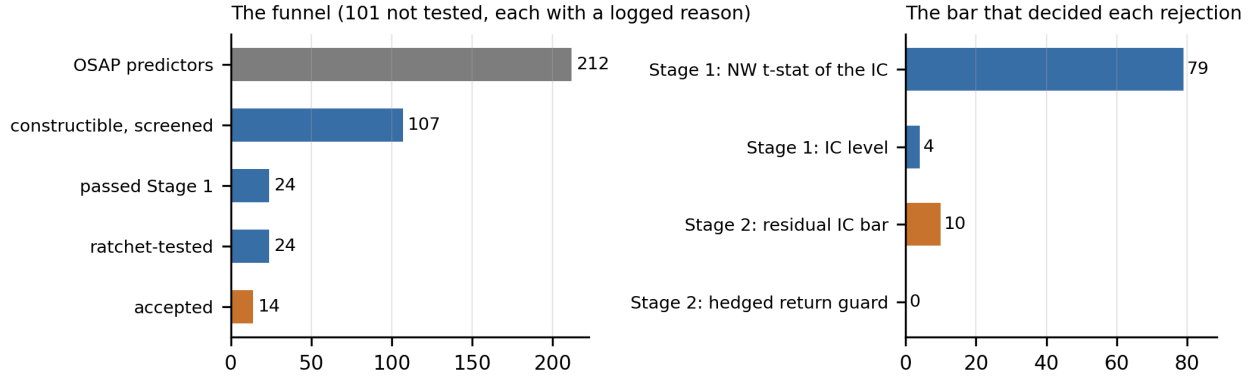


Figure 5. The search funnel and the deciding bars. Left: the search funnel. Every constructible predictor was screened and every Stage 1 passer was ratchet-tested; none was dropped from a ladder. Right: the bar that decided each rejection. At Stage 1 the t-stat bar decides almost everything; at Stage 2 the residual-information bar decides everything and the hedged return guard never bound.

The rejection table carries most of the paper’s evidence. Many well-known signals fail on this universe and window. Most failures are decided by the t-stat bar, and about a third of them point the wrong way. The near misses are visible, so a reader can check what a lower bar would have admitted, and what a raw-return guard that the loop declined to invent would have changed (the raw Δ LS and hedge-part columns of Table AI).

Closing the Search, and What It Cost

The stopping rule was exhaustion of the declared order. The ratchet ran until every Stage 1 passer had a Stage 2 row, which happened at the 24th rung. The search then closed the ratchet, built the construction layer on the frozen composite (Section VIII), asked for and received the data update, reproduced the in-sample composite on the updated data, wrote its expectations for the holdout (Section IX.A), and asked me to declare the search finished and to authorise the out-of-sample evaluation, which I did.

The harness runs took 21.9 hours of compute across 56 completed runs: 6.17 hours of screening, 7.19 of ladders with their baselines and construction runs, and the rest in the construction phase, the data update and the out-of-sample evaluation. The whole search, from the first data download to the out-of-sample evaluation, took three days.

VII. The Selected Factors

Table IX lists the nineteen legs of the final composite: the five baseline characteristics and the fourteen accepted signals, with the catalogue's source for each, what it measures, its standalone IC and t-stat on the decision window, its raw long-short Sharpe and beta, and the two Stage 2 bars it cleared. The Stage 1 columns for the baseline legs are blank because the baseline was never screened; it was measured as a blend. A later stamped diagnostic run prints the standalone within-sector ICs of the baseline legs on this window: 0.0205 (t-stat 3.78) for Profitability, 0.0096 (t-stat 1.27) for Momentum, 0.0058 (t-stat 0.85) for Value, 0.0054 (t-stat 1.21) for Investment and -0.0093 (t-stat -1.53) for Size. Only the profitability leg would have passed the screen on its own. The composite's size, value, investment and momentum families rest on their catalogue standing and on what the ratchet later added to them.

Appendix F describes each accepted leg family by family: what it measures, how it behaved on its own, why it cleared the ratchet, and which near-duplicates it spanned.

Table IX
The Nineteen Legs of v14

Source is the original study as attributed in the Chen and Zimmermann (2022) catalogue at the pinned commit. Stage 1 columns are the standalone screen on 1999 to 2021 (registry rows): the IC and its t-stat, the raw long-short Sharpe and the full-window β of the raw long-short. Residual t-stat and guard t-stat are the Stage 2 bars on the rung that accepted the leg (Table AI). Weight is the leg's share of the composite score in months when every leg scores.

leg	family	since	source	what it measures	Stage 1 IC	Stage 1 t-stat	raw LS Sharpe	β (raw)	residual t-stat	guard t-stat	weight
Size	size	v0	Banz (1981)	log market capitalisation; small is attractive	—	—	—	—	—	—	0.1111
Value	value	v0	Stattman (1980)	book equity / market capitalisation; high is attractive	—	—	—	—	—	—	0.0556
Profitability	profitability	v0	Fama and French (2006)	revenue – COGS – SG&A – interest, / book equity; high is attractive	—	—	—	—	—	—	0.0222
Investment	investment	v0	Cooper, Gulen, and Schill (2008)	year-on-year growth of total assets; low is attractive	—	—	—	—	—	—	0.0556
Momentum	momentum	v0	Jegadeesh and Titman (1993)	12-month return skipping the latest month; high is attractive	—	—	—	—	—	—	0.0556
PctAcc	investment	v1	Hafzalla, Lundholm, and Van Winkle (2011)	(income before extraordinary items – operating cash flow) / [income]; low is attractive	0.0112	4.03	0.76	-0.04	4.67	0.09	0.0556
CBOperProf	profitability	v2	Ball et al. (2016)	cash-based operating profit / assets; high is attractive	0.0209	3.74	0.48	-0.57	3.02	0.95	0.0222
ShareIss5Y	external financing	v3	Daniel and Titman (2006)	5-year growth in split-adjusted shares; low is attractive	0.0158	3.67	0.61	-0.24	3.61	2.83	0.0556

leg	family	since	source	what it measures	Stage 1 IC	Stage 1 t-stat	raw LS Sharpe	β (raw)	residual t-stat	guard t-stat	weight
cfp	value	v4	Desai, Rajgopal, and Venkatachalam (2004)	operating cash flow / market capitalisation; high is attractive	0.0246	3.65	0.62	-0.13	2.97	2.19	0.0556
XFIN	external financing	v5	Bradshaw, Richardson, and Sloan (2006)	(net equity + net debt financing) / assets; low is attractive	0.0170	3.57	0.48	-0.47	3.27	-0.64	0.0556
GP	profitability	v6	Novy-Marx (2013)	(revenue – cost of revenue) / assets; high is attractive	0.0155	3.52	0.69	-0.45	2.74	0.43	0.0222
MaxRet	volatility	v7	Bali, Cakici, and Whitelaw (2011)	maximum daily return in the signal month; low is attractive	0.0233	3.42	0.19	-0.86	4.01	1.75	0.0556
roaq	profitability	v8	Balakrishnan, Bartov, and Faurel (2010)	quarterly net income / prior-quarter assets; high is attractive	0.0208	3.37	0.32	-0.58	3.73	1.95	0.0222
RoE	profitability	v9	Haugen and Baker (1996)	net income / book equity; high is attractive	0.0192	3.34	0.18	-0.59	2.41	-1.26	0.0222
IdioVol3F	volatility	v10	Ang et al. (2006)	one-month volatility of daily three-factor residuals; low is attractive	0.0229	3.15	0.14	-0.98	2.03	0.89	0.0556
STreversal	short-term reversal	v11	Jegadeesh (1990)	last month's return; low is attractive	0.0145	2.96	0.20	0.44	3.81	-0.54	0.1111
zerotrade6M	liquidity	v12	Liu (2006)	zero-volume days and low turnover over six months; illiquid is attractive	0.0185	2.93	0.23	-0.73	2.69	1.72	0.0556
VolumeTrend	liquidity	v13	Haugen and Baker (1996)	5-year trend in share volume, scaled; falling is attractive	0.0114	2.87	0.30	-0.23	2.04	-0.09	0.0556
TrendFactor	momentum	v14	Han, Zhou, and Zhu (2016)	moving-average forecast of next month's return; high is attractive	0.0171	2.79	0.61	0.05	2.14	-0.01	0.0556

Two features of the table deserve a plain statement. First, 7 of the 14 accepted legs have a raw long-short Sharpe below 0.4 despite an IC t-stat above 2.5. The information bar admitted them because they rank the cross-section, and the guard let them through because none of them dragged the hedged blend's return significantly on 276 months. Second, 12 of the 14 have a negative beta on their raw spread, and the composite's beta went from -0.140 to -0.560 . The composite that the rules produced is therefore an information composite whose raw spread is short the market, and the paper's later sections report what the hedge did with that.

VIII. Construction

Stage 3 of the search reports five fixed construction variants on the accepted composite, gross and hedged, for every version: equal-rank deciles, tier-neutral ranking, ICIR-weighted legs, a buffered book and a volatility-targeted book. On v14, on the original data, the equal-rank deciles had the highest Sharpe, 0.983; the buffered book cut long-side turnover from 58.1 to

38.5% a month at a Sharpe of 0.954; tier-neutral ranking (0.858), ICIR weights (0.902) and volatility targeting (0.901) did no better. The ICIR-weighted book's raw beta was -0.93 , the most negative of the five. None of these decides acceptance.

After the ratchet closed, a separate phase built the book an institutional desk would trade. The layer judges nothing: no leg is added, dropped or reweighted by any number it produces, and its parameters were frozen in their own file, stamped as a fifth hash, before its first number. Its alpha is the composite's normal score times the name's predicted volatility. Its gross exposure is $2.0 \times IC_t / 0.02$, floored at zero and capped at 2.0, where IC_t is the composite's trailing 36-month rank IC. The risk model has twelve sector groups plus size and trailing volatility, with exponentially weighted covariances. The decisions file required three changes before the layer's first number (D7). The book is neutral to market beta as well as to sector: the optimiser's constraint matrix stacks the sector dummies and each name's trailing 36-month beta on the same market return the hedge uses, so the target book has zero ex-ante beta. The half-spread comes from a Corwin-Schultz series that the harness builds from daily highs and lows, not from any composite leg. And the regime splits the layer reports are the composite's three best years by the D5 rule, not years chosen by eye. The optimiser is closed-form, followed by name caps, a no-trade buffer and a participation cap of five percent of 21 days' volume per trade. Costs are half the measured spread, a square-root impact term, and 25 basis points a year of borrow. The book is reported at 0.1, 1 and 5 billion dollars from January 2001.

Table X
The Construction Layer on v14

The declared layer row, the fixed-tier spread row and the gross equal-rank deciles at the same AUM, from one continuous run on the updated data (run 057, book January 2001 to September 2026), split into the in-sample period (2001 to 2021) and the holdout (January 2022 to September 2026). Returns in percent per year; turnover one-way percent per month; net β on the universe's cap-weighted return. The measured Corwin-Schultz half-spread averages about 39.2 basis points per unit traded; the fixed schedule (2, 5 and 12 basis points by tier, about 8.3 on average) understates 1999 to 2007. The two rows bracket spending on spread alone.

row	AUM	gross in- sample	gross holdout	net in- sample	net holdout	net Sharpe in- sample	net Sharpe holdout	turnover in- sample	turnover holdout	cost in- sample	cost holdout	net β holdout
layer	\$0.1	4.15	0.74	-0.10	-2.66	-0.02	-0.37	35.2	28.7	4.25	3.40	-0.19
B												

row	AUM	gross in- sample	gross holdout	net in- sample	net holdout	net Sharpe in- sample	net Sharpe holdout	turnover in- sample	turnover holdout	cost in- sample	cost holdout	net β holdout
layer_ fixed_tier_spread	\$0.1 B	4.14	0.75	2.49	-0.40	0.52	-0.05	35.2	28.7	1.65	1.15	-0.19
equal_rank_decile	\$0.1 B	12.81	3.01	-7.62	-14.45	-0.49	-0.60	117.6	107.5	20.43	17.47	-0.83
layer	\$1B	4.16	0.74	-1.69	-3.52	-0.36	-0.49	35.1	28.8	5.86	4.26	-0.19
layer_ fixed_tier_spread	\$1B	4.16	0.74	0.90	-1.26	0.19	-0.17	35.1	28.7	3.26	2.00	-0.19
equal_rank_decile	\$1B	12.81	3.01	-24.39	-24.38	-1.57	-1.01	118.7	108.0	37.21	27.39	-0.83
layer	\$5B	3.95	0.70	-3.71	-5.07	-0.80	-0.70	33.4	28.7	7.65	5.77	-0.18
layer_ fixed_tier_spread	\$5B	3.94	0.73	-1.23	-2.79	-0.27	-0.39	33.4	28.7	5.17	3.52	-0.19
equal_rank_decile	\$5B	12.81	3.01	-56.39	-42.73	-3.33	-1.76	120.9	109.1	69.21	45.75	-0.82

In-sample on the original data, the layer earned 4.08% a year gross at a Sharpe of 0.856 (t-stat 3.27) at 100 million dollars, with costs of 3.39 spread, 0.79 impact and 0.19 borrow, 36.0% one-way monthly turnover, and a net return of -0.29% , a net Sharpe of -0.060 . The net Sharpe fell to -0.409 at one billion and -0.870 at five billion. The beta constraint held: the book's ex-post net beta was -0.101 , against -0.135 for the same book without the constraint. The same composite traded as gross equal-rank deciles earned 12.52% gross on 117.7% turnover and lost after costs, which is the case for a construction layer at all. On the updated data (Table X) the in-sample book restates mildly, to 4.15% gross and a net Sharpe of -0.022 .

On the original data the gross return was front-loaded. It was $+23.6\%$ in 2001 and $+20.3\%$ in 2002; over 2003 to 2020 it compounds to $+37.7\%$ gross, about 1.79% a year, and -36.1% net. The D5 rule removes 2000, 2001 and 2021, and since the book starts in 2001 only two of them, 2001, 2021, fall inside it; without those years the net Sharpe is -0.370 , and over 2011 to 2020 the gross Sharpe is 0.222. Since v14 was selected on this window, these figures describe tradability rather than an expected return. Two further caveats belong with them. The measured spread is the dominant cost, and it is likely overstated for liquid names, because, in the manifest's judgment, the daily Corwin-Schultz estimate is floored at zero before averaging: the fixed-tier spread row, which understates early-sample costs, is net positive in-sample at a Sharpe of 0.502 on the original data (0.519 on the updated data), so the true cost lies between the two rows and no run measures where. And the risk model underpredicts risk by more than a factor of two (on the

original data a bias statistic of 2.58, ex-ante volatility of 1.97% against 4.78% realised; 3.12 over the full book on the updated data): the book is neutral to the modelled sector, size, volatility and beta exposures but positive on every alpha family, and the missing risk is consistent with the families' common factor, which the model omits by design so as not to neutralise the alpha.

IX. The Out-of-Sample Result

A. The Protocol

The protocol, D8, was written before the first screen. It required, in order, a data update once September 2026 had closed, a reproduction of the in-sample composite on the updated data, one continuous run of the composite, the Stage 3 variants and the layer with the out-of-sample months included, with the holdout results read from statistics computed on the holdout months alone, and a holdout-only run as a cross-check. After the out-of-sample evaluation there would be no composite change, no change to the pre-registered specification, no retuning of the layer and no further Stage 2 on this dataset. A new hypothesis needs a new window.

The expectations were written after the in-sample reproduction on the updated data and before any out-of-sample number: the event log records the decision before the first holdout run started. They named, on the updated data, a hedged Sharpe of 0.996 (16.34% a year), a Sharpe without the three best years of 0.666, an excess-of-rf Sharpe of 0.928, a raw Sharpe of 0.790, and a mean IC of 0.0389 over the full window and 0.0283 over its second half, with the instruction to benchmark the holdout IC against the second half. There was no bar: 57 months are below the harness's own 120-month floor, and they can neither confirm nor reject a composite on their own. The decision says so in its text.

B. The Information Coefficient out of Sample

Table XI
The Information Coefficient out of Sample

The composite v14's information coefficient on the holdout against its in-sample baseline on the same data (run 053). The first two holdout rows are run 054's holdout-period statistics, the primary estimate; the others come from

the holdout-only cross-check (run 055), because run 054's holdout-period statistics include no halves, decay or tiers. Decay h1 to h12 is the IC of a score formed one to twelve months before the return month.

measure	holdout, 2022-01..2026-09	in-sample, 1999–2021
mean IC	0.0300	0.0389
IC t-stat (NW)	1.87	5.68
ICIR (holdout-only run)	0.239	0.387
% months IC > 0 (holdout-only run)	63.2	62.3
IC half 1	0.0567	0.0496
IC half 2	0.0043	0.0283
IC decay h1	0.0288	0.0269
IC decay h2	0.0317	0.0226
IC decay h3	0.0235	0.0247
IC decay h6	0.0067	0.0197
IC decay h12	0.0087	0.0235
MEGA IC	0.0202	0.0335
MID IC	0.0404	0.0347
SMALL IC	0.0331	0.0412
avg names per decile	192.8	195.8
months	57	276

The holdout IC is 0.0300 with a Newey-West t-stat of 1.87, positive in 63.2% of months (holdout-only run), 77% of the full in-sample mean and above the second-half benchmark of 0.0283. It is significant at the 10% level but not at 5%: the two-sided p-value is about 0.061 (one-sided 0.030), well short of the t-stat of 2.5 a single predictor needed at Stage 1. On 57 months the level rather than the t-stat is the informative number, and the level held. Its distribution over time did not. By year the IC was +0.105 in 2022, −0.002 in 2023, +0.031 in 2024, −0.021 in 2025 and +0.040 over the nine months of 2026, and the two halves of the holdout differ sharply, 0.0567 in the first and 0.0043 in the second. The mega-cap tier had the weakest IC, as in-sample, and the mid tier the strongest.

C. Returns out of Sample

Table XII
The Long-Short Return out of Sample

The composite v14's gross long-short on the holdout against its in-sample baseline on the same data (run 053). Equal-weight decile 10 minus decile 1 of the family blend, no costs. Holdout rows are run 054's holdout-period statistics unless labelled as from the holdout-only run (055), whose first twelve months run unhedged.

measure	holdout, 2022-01..2026-09	in-sample, 1999–2021
hedged LS Sharpe	0.523	0.996
hedged LS t-stat (NW)	1.18	3.96

measure	holdout, 2022-01..2026-09	in-sample, 1999–2021
hedged LS ann. return %	10.81	16.34
hedged LS ann. vol %	20.65	16.40
hedged LS MaxDD %	-35.6	-43.2
hedged LS hit rate %	59.6	62.7
raw LS Sharpe	0.124	0.790
raw LS ann. return %	3.01	14.91
hedge term, hedged – raw, pp/yr	7.79	1.43
excess-of-rf hedged Sharpe (diagnostic)	0.413	0.928
excess-of-rf hedged ann. return %	8.54	15.09
rf credit, pp/yr	-2.26	-1.25
ex-ante β , mean	-0.549	-0.540
realised β of the raw LS (holdout-only run)	-0.836	-0.556
LS in down markets (market < -3%), %/mo (holdout-only run)	3.62	1.77
down-market months (market < -3%)	15	62
D10 turnover %/mo (holdout-only run)	54.3	58.1
D1 turnover %/mo (holdout-only run)	45.0	54.5
MEGA raw LS Sharpe (holdout-only run)	-0.131	0.275
MID raw LS Sharpe (holdout-only run)	0.126	0.629
SMALL raw LS Sharpe (holdout-only run)	0.209	0.922

The hedged long-short earned 10.81% a year over the 57 months, with a Sharpe of 0.523 and a Newey-West t-stat of 1.18, about half the expected 0.996. Its maximum drawdown, -35.6%, was smaller than the in-sample one. The raw spread earned 3.01% a year, a Sharpe of 0.124, so 7.79 of the 10.81 percentage points were the hedge term, against 1.43 of 16.34 in-sample. Part of the hedge term is mechanical: the excess-of-rf diagnostic removes 2.26 percentage points a year of rf credit, leaving 8.54% a year at a Sharpe of 0.413 (t-stat 0.93).

The rest is the market premium earned by the hedge's long position: the ex-ante beta averaged -0.549 over the holdout, in a market whose trailing twelve-month return was positive in 45 of the 57 months. The hedge was also too small on average. The raw spread's realised beta over those months was -0.836; the ex-ante estimate caught up only late, reaching -1.108 by the final month; in-sample the two agreed (-0.540 ex ante, -0.556 realised). The trailing 36-month estimate under-hedged by about 0.29 of market, so the hedged series stayed net short, and that residual short depressed the hedged return. Both effects are in the headline: the hedged series is neither the raw book nor a neutral one.

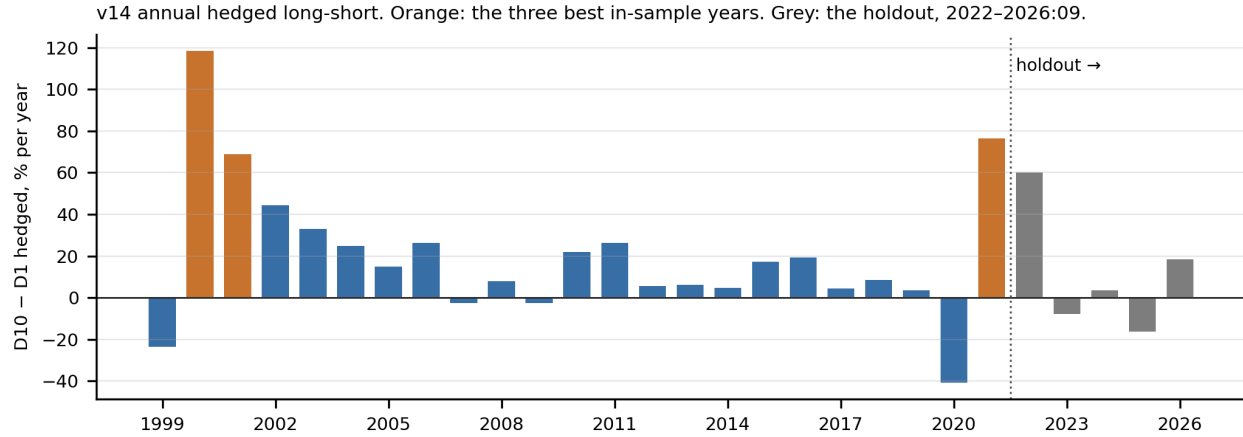


Figure 6. Annual hedged long-short return of v14, in-sample and in the holdout. One continuous run on the updated data (run 056, equal-rank deciles). Orange marks the three best in-sample years that the D5 rule removes (2000, 2001 and 2021); grey is the holdout, whose 2026 bar covers January to September. The holdout's return is carried by 2022.

One year carried the result. The hedged long-short returned +60.0% in 2022, −7.9% in 2023, +3.5% in 2024, −16.2% in 2025 and +18.4% over the nine months of 2026 (Figure 6, from the continuous Stage 3 run). Removing the three best calendar years leaves two of the holdout's five, both negative, so the ex-regime Sharpe (−0.572 for the hedged series in the holdout-only run, −0.708 for the excess series in the primary run) is mechanical here and carries no regime information.

Table XIII
Returns by Decile out of Sample

Average monthly raw return by composite decile, in percent, holdout (run 055, the holdout-only run; run 054's holdout-period statistics include no deciles) against in-sample (run 053).

decile	holdout	in-sample
D1	0.406	0.261
D2	0.632	0.562
D3	0.712	0.761
D4	0.669	0.782
D5	0.630	1.022
D6	0.686	1.067
D7	0.629	1.114
D8	0.649	1.259
D9	0.695	1.350
D10	0.657	1.504
mean of the ten deciles	0.64	0.97

The shape of the deciles changed. In-sample the deciles are monotone from 0.261% a month in decile 1 to 1.504 in decile 10. Out of sample the worst-ranked decile earned 0.406% a month, the least of any decile, and deciles 2 to 10 were flat between 0.629 and 0.712, with the top decile at 0.657, barely above the 0.64 mean of the ten (Figure 7). The ranking still identified the stocks to avoid; it no longer graded the rest. In the 15 holdout months when the market fell by more than three percent the long-short averaged 3.62% a month: the residual short exposure paid when the market fell. By tier, the mega-cap raw spread lost, while the mid and small tiers kept small positive Sharpes.

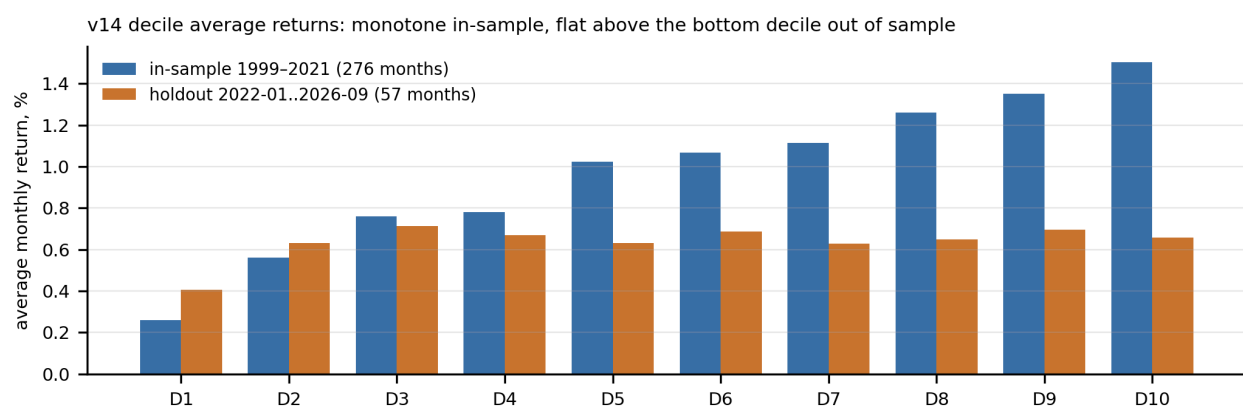


Figure 7. Average monthly return by composite decile, in-sample and in the holdout. In-sample the deciles are monotone; out of sample deciles 2 to 10 are flat and the spread is the bottom decile alone.

The combination of a positive IC and a weak long-short deserves a word. The IC weighs 57 monthly rank correlations equally, over all names. A decile long-short is a bet on the two extremes, and with the top nine deciles flat, the spread is the bottom decile alone. This is a reading rather than a measurement, since no dispersion statistic exists in the results; but the same pattern appears at Stage 1 among the volatility passers, and Section X returns to it.

D. Construction out of Sample

The Stage 3 run printed no separate holdout results, which the protocol had asked for; the gap was found after the out-of-sample evaluation, logged, and not repaired, because the holdout is evaluated only once. Its calendar-year returns show every variant positive in 2022 and negative in 2025. The construction layer at 100 million dollars returned 0.74% a year gross, a Sharpe of

0.101, and -2.66% net after 3.40% of costs, a net Sharpe of -0.367 ; at one and five billion the net Sharpe was -0.485 and -0.701 , and with the fixed-tier spread -0.055 . The layer earned 20.8% gross in 2022 and lost in each of 2023 to 2025. Its net beta on the market was -0.186 in the holdout against -0.098 in-sample: the beta constraint held the book near neutral, and a near-neutral book earned close to nothing gross. That is the cleanest reading of what the composite's information was worth out of sample once the market exposure is removed.

E. Integrity of the Out-of-Sample Test

In the continuous run that includes the holdout, the in-sample portion matches the in-sample run made before the out-of-sample evaluation on all 20 shared statistics to six decimal places, so adding the holdout months changed nothing before them. The construction-layer run's equal-rank decile row reproduces the composite's raw holdout return exactly, at 3.01% a year in both. The holdout-only cross-check begins in January 2022 with no earlier return history, so its beta estimate, which needs 12 months of data, leaves the first twelve months unhedged. It shows a hedged Sharpe of 0.584 , with the same IC and raw return. It also drew the harness's standard warning that 57 months fall below the 120-month minimum, which is expected for a period this short and led to no verdict. No attribution of the holdout result to individual legs or families was run: none was declared in advance, and running one now would mean examining the holdout a second time. The concentration of returns in 2022 and the change in the decile shape are therefore reported without being traced to specific families.

X. Reading the Result

The composite's information was real, and much of it carried out of sample. In-sample its IC averaged 0.0389 , stronger in the first half (0.0496) than in the second (0.0283). Out of sample it was 0.0300 , slightly above the second-half level, so the model held the strength it had shown most recently. As is typical of factor returns, the holdout IC varied from year to year. The ex-regime Sharpe, reported as a diagnostic on every acceptance row, gave a more realistic picture of that variation than the headline Sharpe. A natural extension of the Stage 1 screen, which required

a positive IC in both halves of the window, would be a test of how evenly the information is spread across years.

The information bar did what it was designed to do, and its one side effect is visible throughout the record. Residual IC does not depend on how the portfolio is built, so portfolio choices cannot game it, and it admitted signals that sharpened how well the composite ranked stocks within sectors. Several of these signals, in volatility, illiquidity, profitability and issuance, have a calmer long side than short side. As legs were added, the raw spread therefore leaned short the market, with its beta moving from -0.140 at v_0 to -0.560 at v_{14} . The market hedge was built for exactly this, and in-sample it worked: the ex-ante and realised betas agreed. Out of sample the realised beta deepened to -0.836 in a rising market. A trailing estimate naturally lagged it, so the hedge covered only part of the exposure, and the total-return market proxy credited the short book with interest it did not earn. Both effects are measurable, and each points to a simple refinement. The beta rows are printed on every rung, the guard's reading of the hedged blend is a logged decision, and the risk-free credit is a printed diagnostic. A reader who prefers a beta guard or a headline in excess of the risk-free rate can see exactly which rungs and numbers it would change.

The pattern of a positive IC alongside a weak long-short appears at three scales. At Stage 1, the volatility passers had IC t-statistics near three and raw long-short Sharpes near zero. In the ratchet, MaxRet raised the IC with a significant residual while cutting the raw Sharpe. Out of sample, the composite kept a positive mean IC while its raw spread earned 3.01% a year and its top nine deciles were flat. An IC averages rank agreement over all names and months; a decile long-short is a bet on the extremes. That is a reading, and it was not tested on the holdout.

Economic and empirical families can diverge. Families were assigned by the catalogue's economic label, before any Stage 2 number. `zerotrade6M` is labelled liquidity and opened the liquidity family at a full ninth of the composite. Empirically its annual IC series moves 0.96 with `IdioVol3F`'s and 0.91 with MaxRet's, and its raw spread has a beta of -0.73 : in this universe a low-turnover sort is close to a low-volatility sort. Its admission therefore doubled the composite's

weight on what behaves like one exposure. Families are fixed after Phase C by design, so it stayed. A family structure fitted to the correlation matrix would probably have put it with the volatility legs. That would be a legitimate alternative design, but it is a different pre-registration.

What the search did right is worth stating too. Every number in this paper comes from a stamped result or a record the search wrote itself. The holdout was evaluated once, on a composite and a layer that were both frozen, after a declared data update and a reproduction on the updated data, with its expectations written before the first holdout run started. The mixed result therefore carries evidential weight: it was not chosen, and it is consistent with what the in-sample beta and regime diagnostics had printed.

XI. Limitations

The study rests on one frozen dataset from a single vendor over a single window. Freezing the data makes every result exactly reproducible. Testing the loop on other data sources and periods is the natural next step. The translations are as faithful as Sharadar's fields allow, and where a field departs from the original definition, the choice is declared openly. Depreciation embedded in cost lines is taken as reported rather than re-estimated, operating leases enter debt from the adoption of ASC 842, and every within-sector rank uses the vendor's current sector classification. Performance delistings are rare in this liquid universe, so the Shumway convention for delisting returns seldom applies and has little bearing on the results.

The pre-registration and the holdout each test a specific, well-defined part of the work. The rules and the construction (the family blend, the residual-only Stage 2 bar, the membership band, the within-sector ranks and the hedge) draw on what is known about how factors behaved over 1999 to 2021 and over the years reserved here as the holdout, as any design grounded in the literature does. What the pre-registration secures is the selection: every candidate was judged by rules fixed before it was screened, so no choice inside the search could be tuned to its results. The holdout tests that selection cleanly, and it is not presented as an independent test of the construction. The holdout was protected by rule rather than by withholding data. The first

download already contained those months, and the harness refused any in-search run that reached them, so they were used once, at the end.

Every decision in the loop is recorded together with what was known when it was made. The bars, the window, the lag, the blend, the within-sector ranks and the hedge were all fixed before the first candidate was screened. The families were defined by economic meaning after the Stage 1 results were in, but before any Stage 2 number existed. Rulings on how to apply the rules were logged as each case arose, and the expectations for the holdout were written down after the in-sample results and before the holdout was opened. None of these rulings added a new rule, and when each was written, the holdout was the only data not yet seen.

The hedged headline is not a pure market-neutral return, for two reasons. The hedge uses the market's total return rather than its excess return, which credits the short market position with the risk-free rate, and its trailing beta lagged the holdout's realised beta. Hedging with the market excess return and constraining beta directly in the construction layer correct both, and both corrected returns are small out of sample.

Fifty-seven months is short. The holdout cannot distinguish a dead edge from a long drawdown, one year carries it, and the ex-regime diagnostic is mechanical on five calendar years.

Multiple testing is controlled only by the bar levels. Under a global null, about one Stage 1 false pass (0.984) and about half a Stage 2 false acceptance (0.546) are expected, against 24 passes and 14 acceptances. No family-wise or false-discovery correction was pre-registered.

In-sample layer figures describe tradability only, and the cost range is wide. The two spread rows bracket spending on spread alone, the risk model omits the alpha families' common factor and understates risk by more than a factor of two, and the layer's gross return faded after 2002. About 11% of the in-sample IC arrives on the first trading day of the holding month: starting the forward return at the next month's first trade lowered the composite IC from 0.0389 to 0.0346, mostly in the value, investment and financing legs, with the momentum seed also lower, rather than in the short-horizon legs.

No per-leg or per-family holdout attribution exists, by protocol, so the holdout’s concentration and decile shape are reported without attribution. Families are judgements, declared by label, and one assignment diverged from the data. The seed legs were never screened, and four of the five would not have passed alone.

The loop has been run once, and I designed it. Its autonomy covers the execution of a search, not the choice of research question: the catalogue, the bars, the construction, the window and the phases were mine, fixed before the loop started. A second run on a new catalogue, window or vendor, set up by someone else, would be the real test of whether the framework carries its guarantees without its designer. The run also does not show how the loop behaves with weaker models; the harness would measure the same numbers, but the translation, review and interpretation would likely need more correction.

The agents made mistakes, and the record documents each one. The runner misread figures in its own summaries and overstated several claims in its written reports. Each error was found by a mechanism, a check against the stamped results, the registry or the records checker, and none by the runner noticing it itself. This is why the guarantees in this paper rest on the safeguards enforced by code.

XII. What Transfers

What transfers from this work is the framework: none of its mechanisms depends on the data vendor or on the particular LLM. The composite itself is the least portable part. The data behind it are licensed and cannot be shared, and once its market exposure is removed, its out-of-sample edge was small.

Agents propose, code decides, and a stronger model reviews. The runner ran the loop, the sub-agents did the specialised work, the advisor was consulted at the boundaries and before acceptances, and I resolved five of the seven questions in three decisions. The runner made mistakes, and every one that mattered was caught by a mechanism rather than by a person. That

division of labour is the core of the design, and it is what lets a research programme run for days without supervision and still be audited afterwards.

Pointing the loop at a new problem means changing four things: the catalogue the inventory reads, the data layer and field map the translator writes against, the pre-registration file, and the standing instruction file. The stamps, the phase checkpoints, the hooks, the permission tiers, the record and the roles of the sub-agents stay as they are. The construction used here, ranking within sector and hedging to the market, is set in the pre-registration file and touches none of them. In principle the same loop could search an international-equity or a fixed-income catalogue, or a set of machine-learning signals, under the same guarantees; this paper has shown it on one.

Provenance can be a property of the data and code rather than of anyone's conduct. Stamp the harness, the configuration, the composite and the data on every result, and refuse results that do not match. 56 completed runs later, the pre-registration had provably not moved, and the data had moved once, by a declared and approved data update whose effect on every in-sample number is itself recorded.

A search whose denominator is the whole catalogue, with every exclusion logged and its reason measured, cannot quietly leave the inconvenient predictors unexamined. Inventory everything, then exclude for measured reasons.

Selecting on information and reporting construction separates two questions that are usually confounded. A residual-IC bar cannot be gamed by portfolio choices, and a family blend stops near-duplicates from taking whole votes. This search also shows the bar's limit: it admits legs whose information comes with a market exposure, and it says so in the beta rows rather than in the verdict. Printing the raw series and the beta beside every hedged number is what let the out-of-sample result be read at all.

An order fixed before scores exist, and bars read as registered, are cheap and had teeth. The ratchet's order was declared once and decided between near-twins, and the guard's known bias was disclosed rather than patched mid-search.

Writing the expectations for the holdout before the out-of-sample evaluation is what makes a mixed result reportable rather than a disappointment to be explained away. They were written, the holdout was evaluated once, and the result is reported against them without retuning.

The repository holds every record this paper is built from, the scripts that generate its tables and figures from them, and the commands that re-run any measurement on the dataset. The holdout has been used; re-running it would mean examining it a second time. A new hypothesis needs a new window.

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Appendix A. The Ratchet Ledger and the Families

Table AI
The Stage 2 Ledger

Every Stage 2 rung in declared order, with the two bars and the verdict. The bars are the residual IC t-stat (strictly greater than 2.0) and the paired Δ LS t-stat of the hedged family blend (at least -2.0), both Newey-West with 3 lags. “Base legs” is the number of legs the candidate was projected on: the live composite plus every earlier accepted rung of its ladder. Original data, before the D8 update.

rank	factor	family	Stage 1 t-stat	ladder	base legs	resid IC	residual t-stat (bar > 2.0)	guard t-stat (bar ≥ -2.0)	verdict
1	PctAcc	investment	4.032	L1	5	0.0098	4.674	0.087	accepted
2	CBOperProf	profitability	3.736	L1	6	0.0104	3.020	0.946	accepted
3	ShareIss5Y	external financing	3.667	L1	7	0.0120	3.609	2.833	accepted
4	cfp	value	3.652	L1	8	0.0104	2.974	2.187	accepted
5	XFIN	external financing	3.570	L1	9	0.0069	3.273	-0.637	accepted
6	GP	profitability	3.520	L2	10	0.0073	2.741	0.429	accepted
7	ShareIss1Y	external financing	3.516	L2	11	0.0044	1.879	0.172	rejected
8	MaxRet	volatility	3.424	L2	11	0.0179	4.008	1.749	accepted
9	roaq	profitability	3.366	L2	12	0.0119	3.728	1.952	accepted
10	RoE	profitability	3.337	L2	13	0.0045	2.413	-1.257	accepted
11	OperProfRD	profitability	3.184	L3	14	-0.0011	-0.424	0.924	rejected
12	IdioVol3F	volatility	3.147	L3	14	0.0065	2.031	0.887	accepted
13	NetEquityFinance	external financing	3.028	L3	15	0.0035	1.396	0.473	rejected
14	CF	value	3.028	L3	15	0.0023	1.020	-0.761	rejected
15	STreversal	short-term reversal	2.963	L3	15	0.0153	3.809	-0.536	accepted
16	zerotrade6M	liquidity	2.927	L4	16	0.0099	2.687	1.721	accepted
17	VolumeTrend	liquidity	2.870	L4	17	0.0047	2.044	-0.087	accepted
18	zerotrade12M	liquidity	2.832	L4	18	0.0010	0.490	0.136	rejected
19	RealizedVol	volatility	2.810	L4	18	0.0055	1.730	-0.612	rejected
20	TrendFactor	momentum	2.787	L4	18	0.0078	2.141	-0.005	accepted
21	BidAskSpreadFlip	liquidity	2.763	L5	19	0.0051	1.769	0.427	rejected
22	IdioVolAHT	volatility	2.628	L5	19	0.0073	1.924	0.062	rejected
23	zerotrade1M	liquidity	2.626	L5	19	-0.0025	-1.278	0.114	rejected
24	NetPayoutYield	value	2.605	L5	19	0.0003	0.116	0.060	rejected

Table AI (continued)
Diagnostics on Every Rung

The diagnostics printed on every rung and never used as bars: the composite’s hedged long-short Sharpe without and with the candidate; the change in the blend’s raw long-short return and the hedge part of the change in its hedged return (hedged change minus raw change, pp per year), whose sum is the change the guard reads; the change in the full-window β of the raw long-short; the paired composite Δ IC t-stat; the spanning alpha t-stat of the candidate’s long-short on the composite’s; and the share of the candidate’s IC that survives projection on the legs. Computed from the registry’s with and without arms.

rank	factor	verdict	Sharpe without	Sharpe with	Δ Sharpe	raw Δ LS	hedge part	$\Delta\beta$	paired Δ IC t-stat	spanning alpha t-stat	residual share of IC
1	PctAcc	accepted	0.600	0.614	+0.014	-0.01	+0.07	+0.019	2.81	2.47	0.88
2	CBOperProf	accepted	0.614	0.661	+0.047	+0.31	+0.13	-0.001	-0.01	3.71	0.50

rank	factor	verdict	Sharpe without	Sharpe with	Δ Sharpe	raw ALS	hedge part	$\Delta\beta$	paired AIC t-stat	spanning alpha t-stat	residual share of IC
3	ShareIss5Y	accepted	0.661	0.848	+0.187	+1.46	+1.01	-0.070	3.03	4.51	0.76
4	cfp	accepted	0.848	0.996	+0.148	+0.92	+0.81	-0.071	3.10	0.40	0.42
5	XFIN	accepted	0.996	0.914	-0.082	-0.41	-0.05	-0.082	-0.62	1.28	0.41
6	GP	accepted	0.914	0.925	+0.011	+0.35	-0.20	+0.022	-0.79	3.62	0.47
7	ShareIss1Y	rejected	0.925	0.900	-0.026	+0.04	+0.03	-0.005	1.76	2.61	0.26
8	MaxRet	accepted	0.925	0.951	+0.026	-0.59	+2.82	-0.327	2.61	1.38	0.77
9	roaq	accepted	0.951	0.967	+0.016	+0.72	+0.51	-0.072	2.66	0.39	0.57
10	RoE	accepted	0.967	0.932	-0.036	-0.45	+0.14	-0.013	1.44	-0.33	0.24
11	OperProfRD	rejected	0.932	0.944	+0.013	+0.19	-0.02	-0.002	-0.20	0.86	-0.05
12	IdioVol3F	accepted	0.932	0.919	-0.012	+0.41	-0.03	-0.018	0.81	-0.85	0.28
13	NetEquityFinance	rejected	0.919	0.908	-0.011	+0.01	+0.18	-0.026	1.86	-0.28	0.21
14	CF	rejected	0.919	0.861	-0.058	-0.67	+0.28	-0.039	0.72	-2.70	0.11
15	STreversal	accepted	0.919	0.942	+0.023	+1.09	-1.73	+0.240	1.50	-0.48	1.06
16	zerotrade6M	accepted	0.942	0.985	+0.042	+0.64	+0.90	-0.123	0.99	0.76	0.53
17	VolumeTrend	accepted	0.985	0.979	-0.006	+0.18	-0.21	+0.015	-0.19	0.67	0.42
18	zerotrade12M	rejected	0.979	0.985	+0.007	-0.06	+0.11	-0.006	0.10	0.79	0.06
19	RealizedVol	rejected	0.979	0.956	-0.023	-0.38	+0.25	-0.025	0.37	-0.50	0.23
20	TrendFactor	accepted	0.979	0.983	+0.004	+0.77	-0.77	+0.060	1.36	1.77	0.46
21	BidAskSpreadFlip	rejected	0.983	0.951	-0.031	-0.20	+0.40	-0.057	1.83	-0.73	0.25
22	IdioVolAHT	rejected	0.983	0.967	-0.016	+0.02	+0.00	-0.025	0.71	-0.69	0.31
23	zerotrade1M	rejected	0.983	0.986	+0.004	-0.03	+0.08	-0.019	-0.36	-0.74	-0.16
24	NetPayoutYield	rejected	0.983	0.982	+0.000	-0.43	+0.45	-0.033	-0.56	-1.05	0.02

Table AII
The Declared Families

The nine declared families, their definitions, every member in assignment order with the catalogue's economic label and the Stage 2 verdict, and the live legs in v14 with the family's weight (research/families.yaml).

family	definition	members (assignment order; catalogue label; verdict)	live legs in v14	weight in v14
size	market capitalisation (small = attractive)	Size (v0 seed)	Size	0.1111
value	price relative to a fundamental anchor	Value (v0 seed); CF [valuation] (rejected); NetPayoutYield [valuation] (rejected); cfp [valuation] (accepted)	Value, cfp	0.1111
profitability	earning power relative to capital	Profitability (v0 seed); CBOperProf [profitability] (accepted); GP [profitability] (accepted); OperProfRD [profitability] (rejected); RoE [profitability] (accepted); roaq [profitability] (accepted)	Profitability, CBOperProf, GP, roaq, RoE	0.1111
investment	growth of the asset base or of investment	Investment (v0 seed); PctAcc [accruals] (accepted)	Investment, PctAcc	0.1111
momentum	continuation of past returns	Momentum (v0 seed); TrendFactor [momentum] (accepted)	Momentum, TrendFactor	0.1111
external financing	net capital raised from or returned to investors (share issuance, net equity and debt financing)	NetEquityFinance [external financing] (rejected); ShareIss1Y [external financing] (rejected); ShareIss5Y [external financing] (accepted); XFIN [external financing] (accepted)	ShareIss5Y, XFIN	0.1111
volatility	dispersion or tail size of a stock's own returns (total, idiosyncratic, extreme daily); low = attractive	IdioVol3F [volatility] (accepted); IdioVolAHT [volatility] (rejected); MaxRet [volatility] (accepted); RealizedVol [volatility] (rejected)	MaxRet, IdioVol3F	0.1111

family	definition	members (assignment order; catalogue label; verdict)	live legs in v14	weight in v14
short-term reversal	reversal of the most recent month's return	STreversal [short-term reversal] (accepted)	STreversal	0.1111
liquidity	trading activity and trading cost (turnover, zero-volume days, bid-ask spread, volume trend)	VolumeTrend [volume] (accepted); zero-trade12M [liquidity] (rejected); zero-trade1M [liquidity] (rejected); zero-trade6M [liquidity] (accepted); BidAskSpreadFlip [liquidity] (rejected)	zero-trade6M, VolumeTrend	0.1111

Appendix B. Stage 1 Rejections

Table BI
Stage 1 Rejections

Every predictor rejected at the standalone screen, with its mean rank IC, Newey-West t-stat, half-window ICs, raw long-short Sharpe and annual return, the full-window β of the raw long-short, coverage, and the bar that decided it. Bars: IC ≥ 0.010 ; NW t-stat ≥ 2.50 ; IC > 0 in both halves; raw gross D10–D1 return > 0 ; coverage $\geq 40\%$; ≥ 30 names per decile. The decided-by column names the t-stat bar when it fails, else the first failed bar.

factor	batch	mean IC	IC		raw LS Sharpe	raw LS ret %	β (raw)	coverage %	decisive bar	flip note
			t-stat (NW)	IC half 1	IC half 2					
Accruals	B1	0.0048	1.84	0.0075	0.0021	0.402	2.92	0.04	76.7	NW t-stat
AM	B1	0.0098	1.26	0.0168	0.0028	0.216	4.72	0.34	99.6	NW t-stat
AnnouncementReturn	B1	0.0092	2.28	0.0102	0.0081	0.296	2.51	-0.21	66.5	NW t-stat
Beta	B1	-0.0091	-0.97	-0.0075	-0.0106	0.049	1.12	1.11	85.4	NW t-stat
BetaFP	B1	-0.0123	-1.28	-0.0144	-0.0103	0.047	1.13	1.22	83.1	NW t-stat
BetaLiquidityPS	B1	-0.0008	-0.27	0.0013	-0.0029	-0.076	-0.57	0.11	67.2	NW t-stat
BetaTailRisk	B1	-0.0006	-0.06	-0.0004	-0.0008	0.069	1.09	0.78	60.2	NW t-stat
BidAskSpread	B1	-0.0205	-2.77	-0.0195	-0.0215	-0.041	-0.94	0.97	99.1	NW t-stat
										reversal declared and screened as BidAskSpreadFlip
BMdec	B1	0.0045	0.82	0.0118	-0.0028	0.311	3.68	-0.03	92.1	NW t-stat
BookLeverage	B1	-0.0096	-2.59	-0.0072	-0.0120	-0.348	-3.63	-0.04	99.6	NW t-stat
BPEBM	B1	-0.0092	-2.06	-0.0072	-0.0112	0.034	0.39	0.23	75.6	NW t-stat
Cash	B2	-0.0091	-1.86	-0.0091	-0.0092	-0.036	-0.57	0.33	99.6	NW t-stat
CashProd	B2	0.0030	0.47	0.0094	-0.0035	0.259	3.74	0.26	99.2	NW t-stat
ChAssetTurnover	B2	0.0058	2.24	0.0084	0.0033	0.380	2.37	-0.06	83.4	NW t-stat
ChEQ	B2	0.0057	1.40	0.0068	0.0047	0.692	7.66	0.02	91.2	NW t-stat
ChInv	B2	0.0052	1.51	0.0093	0.0010	0.602	5.14	-0.10	60.3	NW t-stat
ChInvIA	B2	0.0009	0.36	0.0045	-0.0026	0.591	4.04	-0.11	87.2	NW t-stat
ChNNCOA	B2	0.0094	4.15	0.0146	0.0041	0.697	4.81	-0.07	77.2	IC level

factor	batch	mean IC	IC t-stat (NW)	IC half 1	IC half 2	raw LS Sharpe	raw LS ret %	β (raw)	coverage %	decisive bar	flip note
ChNWC	B2	0.0042	2.04	0.0049	0.0035	0.194	1.15	-0.02	77.0	NW t-stat	
ChTax	B2	0.0048	1.42	0.0042	0.0055	0.012	0.09	-0.03	87.1	NW t-stat	
CompEquIss	B2	0.0119	2.44	0.0134	0.0104	0.428	3.62	-0.06	65.1	NW t-stat	
CompositeDebtIssuance	B2	0.0047	1.49	0.0102	-0.0008	0.153	1.33	-0.11	46.2	NW t-stat	
CoskewACX	B3	0.0062	1.65	-0.0011	0.0135	0.256	2.78	0.22	91.1	NW t-stat	
Coskewness	B3	0.0068	1.65	-0.0001	0.0137	0.130	1.61	0.33	91.1	NW t-stat	
DelCOA	B3	0.0009	0.26	0.0074	-0.0055	0.612	4.97	-0.07	77.3	NW t-stat	
DelCOL	B3	-0.0035	-0.92	0.0025	-0.0095	0.090	0.81	-0.06	77.1	NW t-stat	
DelEqu	B3	0.0052	1.25	0.0060	0.0045	0.580	6.06	0.00	95.2	NW t-stat	
DelFINL	B3	0.0039	1.36	0.0104	-0.0026	0.277	2.26	-0.18	69.6	NW t-stat	
DelNetFin	B3	0.0025	1.08	0.0074	-0.0024	0.154	1.07	0.00	77.2	NW t-stat	
dNoa	B9	0.0077	2.19	0.0156	-0.0002	0.947	7.74	-0.07	95.2	NW t-stat	
DolVol	B3	-0.0008	-0.18	0.0056	-0.0073	0.364	3.69	-0.01	99.5	NW t-stat	
EarningsConsistency	B3	0.0090	2.60	0.0055	0.0125	-0.014	-0.13	-0.18	43.3	IC level	
EarningsSurprise	B3	0.0048	1.63	0.0023	0.0072	0.190	1.33	-0.09	86.2	NW t-stat	
EBM	B3	0.0018	0.42	0.0125	-0.0088	0.320	3.22	0.24	75.6	NW t-stat	
EntMult	B4	0.0134	1.96	0.0320	-0.0052	0.371	5.55	-0.08	65.9	NW t-stat	
EP	B3	0.0055	0.96	0.0135	-0.0026	0.333	3.53	-0.05	75.1	NW t-stat	
EquityDuration	B4	0.0104	1.86	0.0203	0.0004	0.181	2.33	-0.30	94.7	NW t-stat	
grcapx	B9	0.0041	1.37	0.0067	0.0015	0.555	4.02	-0.08	81.0	NW t-stat	
grcapx3y	B9	0.0028	0.89	0.0044	0.0012	0.523	3.97	-0.04	72.3	NW t-stat	
GrLTNOA	B4	-0.0059	-2.75	-0.0049	-0.0069	-0.013	-0.08	0.04	76.8	NW t-stat	reversal qualified; not screened (reversed IC below 0.010)
GrSaleToGrInv	B4	0.0035	1.58	0.0065	0.0005	0.233	1.62	-0.01	57.1	NW t-stat	
GrSaleToGrOverhead	B4	-0.0020	-0.75	-0.0002	-0.0038	-0.447	-3.04	-0.04	85.1	NW t-stat	
Herf	B4	-0.0028	-1.15	-0.0027	-0.0029	0.004	0.03	0.04	91.5	NW t-stat	
HerfAsset	B4	-0.0049	-2.00	-0.0037	-0.0062	-0.075	-0.59	0.06	91.6	NW t-stat	
HerfBE	B4	-0.0030	-1.21	-0.0005	-0.0056	0.039	0.28	-0.03	91.6	NW t-stat	
High52	B4	0.0065	0.77	0.0011	0.0120	-0.005	-0.13	-1.01	96.1	NW t-stat	
Illiquidity	B5	-0.0064	-1.19	0.0000	-0.0127	0.294	4.31	0.29	96.4	NW t-stat	

factor	batch	mean IC	IC t-stat (NW)	IC half 1	IC half 2	raw LS Sharpe	raw LS ret %	β (raw)	coverage %	decisive bar	flip note
IntanBM	B5	-0.0003	-0.04	0.0034	-0.0040	0.259	4.34	0.56	61.1	NW t-stat	
IntanCFP	B5	-0.0033	-0.40	-0.0013	-0.0053	0.254	4.67	0.65	62.9	NW t-stat	
IntanEP	B5	-0.0045	-0.55	-0.0023	-0.0067	0.248	4.58	0.64	62.9	NW t-stat	
IntanSP	B5	-0.0052	-0.62	-0.0023	-0.0081	0.275	5.20	0.67	62.9	NW t-stat	
IntMom	B5	0.0031	0.50	-0.0011	0.0073	-0.077	-1.12	-0.18	95.4	NW t-stat	
InvestmentTWX	B5	0.0003	0.15	0.0000	0.0006	0.480	2.68	0.03	78.7	NW t-stat	
InvGrowth	B5	0.0055	1.56	0.0094	0.0016	0.352	3.50	-0.14	51.7	NW t-stat	
Leverage	B5	0.0125	1.71	0.0168	0.0082	0.344	7.09	0.30	99.6	NW t-stat	
LRreversal	B5	-0.0080	-1.29	-0.0036	-0.0123	0.215	2.84	0.09	78.9	NW t-stat	
MeanRankRevGrowth	B6	-0.0005	-0.12	0.0019	-0.0029	0.272	2.23	0.10	46.6	NW t-stat	
Mom12mOffSeason	B6	0.0084	1.24	0.0047	0.0121	0.314	7.08	-0.35	96.4	NW t-stat	
Mom6m	B6	0.0077	1.24	0.0060	0.0094	0.303	6.47	-0.47	98.2	NW t-stat	
MomOffSeason	B6	-0.0052	-0.97	0.0015	-0.0119	0.213	2.23	0.05	66.6	NW t-stat	
MomOffSeason06YrPlus	B6	0.0065	1.76	0.0079	0.0052	0.244	1.45	0.01	40.6	NW t-stat	
MomSeason	B6	0.0020	0.58	0.0090	-0.0049	-0.038	-0.27	0.00	66.1	NW t-stat	
MomSeason06YrPlus	B6	0.0069	1.90	0.0124	0.0015	0.030	0.21	0.00	40.3	NW t-stat	
MomSeasonShort	B6	0.0013	0.28	0.0054	-0.0028	-0.042	-0.44	0.02	96.1	NW t-stat	
MRreversal	B5	-0.0043	-0.96	-0.0028	-0.0057	0.314	3.69	0.09	91.0	NW t-stat	
NetDebtFinance	B6	0.0034	1.38	0.0087	-0.0019	0.321	2.25	-0.16	88.1	NW t-stat	
NOA	B6	0.0016	0.62	0.0052	-0.0019	1.008	8.64	-0.02	77.3	NW t-stat	
OPLeverage	B7	0.0087	2.89	0.0101	0.0073	0.505	4.09	-0.10	85.4	IC level	
OrgCap	B7	0.0054	1.70	0.0043	0.0065	0.399	4.21	0.08	50.4	NW t-stat	
PctTotAcc	B7	0.0022	0.91	0.0014	0.0029	0.434	2.48	-0.08	95.6	NW t-stat	
Price	B7	-0.0100	-1.41	-0.0027	-0.0174	0.137	2.98	0.77	100.0	NW t-stat	
PriceDelayRsqr	B7	-0.0070	-2.12	-0.0009	-0.0131	-0.122	-1.09	-0.17	87.8	NW t-stat	
PriceDelaySlope	B7	-0.0055	-1.69	-0.0069	-0.0042	-0.080	-0.60	0.23	87.8	NW t-stat	
PriceDelayTstat	B7	-0.0055	-1.68	-0.0069	-0.0042	-0.078	-0.59	0.23	87.8	NW t-stat	
ResidualMomentum	B7	0.0035	0.72	0.0062	0.0009	0.162	1.76	-0.24	67.4	NW t-stat	
ReturnSkew	B7	0.0050	2.22	0.0074	0.0026	0.061	0.40	-0.03	99.8	NW t-stat	

factor	batch	mean IC	IC t-stat (NW)	IC half 1	IC half 2	raw LS Sharpe	raw LS ret %	β (raw)	coverage %	decisive bar	flip note
ReturnSkew3F	B8	0.0007	0.35	0.0022	-0.0007	-0.223	-1.34	0.02	96.7	NW t-stat	
RevenueSurprise	B8	0.0094	2.78	0.0059	0.0129	0.250	1.82	-0.05	86.6	IC level	
SP	B8	0.0182	2.50	0.0270	0.0093	0.468	9.21	0.18	96.1	NW t-stat	
TotalAccruals	B8	0.0000	-0.01	-0.0004	0.0003	0.191	1.47	0.12	94.6	NW t-stat	
VarCF	B8	0.0064	0.76	0.0017	0.0110	-0.226	-3.97	-0.79	60.7	NW t-stat	
VolMkt	B8	0.0161	2.25	0.0135	0.0187	0.105	2.38	-0.94	97.1	NW t-stat	
VolSD	B8	0.0112	2.36	0.0147	0.0078	0.151	1.95	-0.47	79.8	NW t-stat	

Appendix C. Predictors Not Tested

Table CI
Predictors Not Screened

The 101 OSAP predictors (Chen and Zimmermann (2022)) not screened, grouped by class, each with the measured or structural reason recorded at the inventory (osap_source/osap_frontier.yaml, text shortened where marked). “Data unavailable” means an input Sharadar does not publish; “pre-screen checks” means a coverage, mass-point, discrete-value or binary-indicator failure measured on this dataset; “data start” means the data begin too late for the 120-month minimum.

predictor	class	reason recorded at the inventory
AbnormalAccruals	data unavailable	needs compustat.ppeg (gross PP&E) as a Jones-model regressor; SF1 has only ppenet; OSAP does not zero-fill ppeg, so the rule makes it infeasible ...
AccrualsBM	data unavailable	needs compustat.txp (income taxes payable); no SF1 column (112 checked; taxliabilities is total tax liabilities); AccrualsBM uses raw txp and txp is ...
Activism1	data unavailable	GIM governance index G (external spreadsheet, ends 2007-01, no Sharadar source) and 13F maxinstown_perc (SF3 from 2013-06-30): supports disjoint, ...
Activism2	data unavailable	GIM governance index G (external, ends 2007-01, no Sharadar source) and 13F largest-holder share (SF3 from 2013-06-30): supports disjoint, zero ...
AdExp	data unavailable	needs compustat.xad (advertising expense); no SF1 column among 112 (sgna/opex are aggregates); AdExp uses raw xad, not zero-filled, so no approx route
AgeIPO	data unavailable	needs FoundingYear and IPO dates from Ritter's external IPO-age file (PERMNO-keyed); no Sharadar field or crosswalk; not zero-filled. ...
AnalystRevision	data unavailable	only input is IBES consensus meanest (fpi 1); no Sharadar table carries estimates; not a Compustat item, no zero-fill
AnalystValue	data unavailable	requires IBES analyst forecasts (feps1, feps2, LTG); no Sharadar table carries estimates; the source drops firms without forecasts (no zero-fill)
AOP	data unavailable	requires IBES analyst forecasts (feps1, feps2, LTG); no Sharadar table carries estimates; the source drops firms without forecasts (no zero-fill)
betaVIX	data unavailable	needs daily VIX (or VXO) changes; ^VIX exists only under Sharadar table SFP, which is mapped but not held (adding it moves DATA_SHA, stop-and-ask 3); ...
BrandInvest	data unavailable	needs compustat.xad (advertising) for BrandCapital; no SF1 column; xad is not in OSAP zero_fill_vars and BrandCapital is NaN where xad is NaN (xad0 ...
ChangeInRecommendation	data unavailable	IBES analyst recommendation code (ireccd); no Sharadar recommendations/estimates table; no zero-fill; no proxy
ChForecastAccrual	data unavailable	sign of the monthly change in IBES consensus EPS (meanest, fpi 1); no Sharadar estimates table; not zero-filled (NaN without it); the accruals filter ...
ChNAnalyst	data unavailable	change in IBES analyst count (numest, fpi 1); no Sharadar estimates table; NaN when numest or its lag is missing (no zero-fill)
CitationsRD	data unavailable	numerator is NBER patent citations (ncitscale), not published by Sharadar; OSAP's fillna(0) fills gaps in that panel, so dropping it makes the signal ...

predictor	class	reason recorded at the inventory
ConsRecomm	data unavailable	mean IBES recommendation code (ireccd); no Sharadar recommendations table; no zero-fill or optional term
ConvDebt	data unavailable	needs convertible debt (dc from dcvt/dcpstk) and cshrc; neither in any held table; OSAP zero-fills dc but not cshrc; dropping both leaves a constant 0
CPVolSpread	data unavailable	OptionMetrics call minus put implied volatility; Sharadar publishes no options data; OSAP drops missing rows (no zero-fill)
CredRatDG	data unavailable	needs S&P credrat / Capital IQ downgrade history; Sharadar has no ratings column or EVENTS proxy; OSAP's fillna(0) fills no-downgrade months, not an ...
CustomerMomentum	data unavailable	needs Compustat Segment customer names and the CCM link; the customer link is the signal; not in Sharadar
dCPVolSpread	data unavailable	needs OptionMetrics call and put implied volatility; no held Sharadar table carries option data; OSAP leaves NaN (no zero-fill)
DebtIssuance	data unavailable	needs gross long-term debt issuance (dltis); SF1 has only net ncfdt (a different event: net borrower incl. commercial paper); dltis not in OSAP ...
DownRecomm	data unavailable	needs IBES recommendation history (downgrades); no Sharadar table carries analyst recommendations; not an OSAP zero-fill item; 0 of 276 months ...
dVolCall	data unavailable	needs OptionMetrics 30-day 50-delta call implied volatility; no held Sharadar table carries option data (SF3A call holdings are 13F positions); OSAP ...
dVolPut	data unavailable	needs OptionMetrics 30-day 50-delta put implied volatility; not in Sharadar; OSAP does not zero-fill it
EarningsForecastDisparity	data unavailable	every input is IBES (meanest FY1, long-term growth fgr5yr, fy0a); no analyst data in Sharadar; not zero-filled by OSAP
EarningsStreak	data unavailable	inputs actual, meanest, price, anndats_act, statpers from IBES_EPS_Adj (analyst surprise streak); not in Sharadar; not zero-filled by OSAP; an SF1 ...
ExclExp	data unavailable	signal is int0a - epspiq; int0a is the IBES unadjusted actual EPS, not in Sharadar and not zero-filled by OSAP (missing -> row dropped); epspiq alone ...
FEPS	data unavailable	signal is IBES meanest (mean unadjusted FY1 EPS forecast); no analyst forecasts in Sharadar; not zero-filled by OSAP
fgr5yrLag	data unavailable	needs the IBES long-term growth forecast (meanest, fpi 0); no analyst data in Sharadar; OSAP requires it non-missing
ForecastDispersion	data unavailable	signal is IBES stdev / meanest (FY1 forecast dispersion); no analyst data in Sharadar; OSAP does not zero-fill it
FR	data unavailable	needs Compustat pension items (pbnaa, pplao, pplau, pbnv, pbpro, pbpru; COMP.ACO_PNFNDA); SF1 has no pension field; OSAP inner-merges (no ...
Governance	data unavailable	G-index (Gompers-Ishii-Metrick, Yale spreadsheet) is in no Sharadar table; OSAP drops rows without G (no default); OSAP's own series also ends ...
GrAdExp	data unavailable	advertising expense xad is the signal itself; no Sharadar field; OSAP reads raw xad (not zero-filled) and nulls xad < 0.1, so no zero-fill rescue
hire	data unavailable	employee counts (Compustat emp) are in no held Sharadar table; OSAP sets hire = 0 where emp is missing, so without emp the signal is the constant 0 ...
InvestPPEInv	data unavailable	needs gross PP&E (ppeg), which SF1 does not carry (only ppenet) and OSAP does not zero-fill; substituting net PP&E is a different signal and is not ...
IO_ShortInterest	data unavailable	short interest is in no held Sharadar table (13 checked), and OSAP uses it as the sample filter (top 1% of shortint/shrout), so without it every row ...
iomom_cust	data unavailable	needs BEA Make input-output tables and Compustat NAICS to map firms to BEA industries; neither is in the dataset; OSAP has no fallback and does not ...
iomom_supp	data unavailable	needs BEA Use input-output tables and Compustat NAICS; neither is in the dataset; OSAP has no fallback and does not zero-fill
Mom6mJunk	data unavailable	needs an S&P/CIQ credit rating (spltrcm, comp.adsprate, CIQ fallback) to select junk-rated firms; no held Sharadar table carries ratings (EVENTS ...
NetDebtPrice	data unavailable	numerator (dltt + dlc + pstk + dvpa - tstkp) - che needs preferred stock pstk; no SF1 field, and neither OSAP's zero_fill_vars nor NetDebtPrice's ...
OptionVolume1	data unavailable	numerator is OptionMetrics option volume (optvolume); no held Sharadar table carries option volume (SF3A/SF3B put/call are quarterly 13F holdings ...
OptionVolume2	data unavailable	OptionMetrics option volume (optvolume) and its 6-month average; no held Sharadar table carries option data; OSAP does not zero-fill it
OrderBacklog	data unavailable	order backlog (Compustat ob) has no SF1 field; OSAP zero-fills ob but its predictor then sets the signal NaN wherever ob == 0, so with ob absent the ...

predictor	class	reason recorded at the inventory
OrderBacklogChg	data unavailable	needs non-zero order backlog (Compustat ob) at t and t-12; ob has no SF1 field and OSAP's zero-fill-then-null logic makes the signal undefined for ...
PatentsRD	data unavailable	numerator npat comes from a patent panel Sharadar does not publish; the predictor's fillna(0) fills gaps inside that panel, so without it npat is 0 ...
PayoutYield	data unavailable	numerator needs preferred redemption value pstkrv (no SF1 field, not zero-filled by OSAP or the predictor) and gross repurchases prstk (Sharadar has ...)
PredictedFE	data unavailable	needs IBES forecasts (feps1, feps2, long-term growth) and AOP (itself built from IBES) as regressors; no analyst data in Sharadar; the source screen ...
ProbInformedTrading	data unavailable	PIN parameters (a, u, es, eb) come from a third-party file estimated on intraday buy/sell trade counts; no held Sharadar table carries trade counts ...
RDS	data unavailable	OSAP zero-fills recta, msa and pension terms but then sets RDS NaN wherever recta and msa were originally missing (both_missing_mask); SF1 carries ...
realestate	data unavailable	needs Compustat fatb and fatl (buildings and land at cost) and ppeg, with a ppenb/ppenls fallback; none is in SF1 and OSAP zero-fills none (a ...)
Recomm_ShortInterest	data unavailable	needs IBES analyst recommendations and Compustat short interest; neither in Sharadar; OSAP inner-joins both (no zero-fill); also a binary flag in two ...
retConglomerate	data unavailable	needs Compustat business-segment data (segment SICs and sales) and the CCM link; no held Sharadar table carries segments; OSAP inner-merges (no fill)
REV6	data unavailable	needs IBES mean-estimate revisions (meanest, statpers, fpedats); no analyst data in Sharadar; NaN if any of the 7 monthly changes is missing (no ...)
RIO_Dis	data unavailable	needs IBES forecast dispersion (stdev, not in Sharadar, no zero-fill) and 13F ownership from 2013-06 (94 scorable months < 120); 5-valued output
RIVolSpread	data unavailable	needs at-the-money implied volatility (OptionMetrics); no held Sharadar table carries option data; OSAP keeps only rows with implied vol (no ...)
sfe	data unavailable	needs the IBES median next-year EPS forecast and analyst count; no analyst data in Sharadar; OSAP inner-merges (no fill)
ShortInterest	data unavailable	short interest (shortint) is the signal itself and is in no held Sharadar table; OSAP inner-merges on the short-interest file (no zero-fill)
sinAlgo	data unavailable	needs Compustat segments and historical NAICS (neither in Sharadar; OSAP does not zero-fill them); a SIC-only build is a binary flag scored only in a ...
skew1	data unavailable	built from OptionMetrics option-level implied volatility; no held Sharadar table carries option data; OSAP does not zero-fill it
SmileSlope	data unavailable	needs OptionMetrics implied-volatility surface (30-day, delta 50 put minus call); no held Sharadar table carries option data; OSAP keeps only ...
tang	data unavailable	the pinned code uses gross PP&E (ppeg), not in SF1 and dropped by OSAP when missing (ppnet is a substitution, not adopted); and the ...
Tax	data unavailable	needs Compustat txfo, txfed, txdi (no SF1 fields, not zero-filled by OSAP); the fallback txt - txdi is NaN without txdi, so every row is NaN ...
UpRecomm	data unavailable	needs IBES analyst recommendations (ireccd); no analyst data in Sharadar; also a binary upgrade flag
DelDRC	pre-screen checks	coverage: 9.7% / 27.4% / 28.0% of the universe at the 1998-12 / 2010-06 / 2021-11 probes, under the 40% Stage 1 bar at every probe; structural, from ...
DelITI	pre-screen checks	mass point: change in long-term investments (investmentsnc) is exactly 0 for 50.7-59.9% of scored names (40-46% of the universe; OSAP's ivao ...)
DivInit	pre-screen checks	binary dividend-initiation flag: 98.7% of the universe is 0 (24.7 ones per month on average; 181 of 263 usable months have < 30 ones), so qcut gives ...
DivOmit	pre-screen checks	binary dividend-omission flag: 99.6% of the universe is 0 (7.2 ones per month on average, 206 of 265 months < 10, 3 months with none); mode breaches ...
DivSeason	pre-screen checks	binary seasonal-dividend flag: 71.5-78.7% of scored names are 0 (scored = payers within 12 months, ~52% of the universe), so qcut gives 2 bins and ...
DivYieldST	pre-screen checks	predicted-dividend-yield tercile code in {0,1,2,3}: 76.0-83.5% of scored names are 0, so qcut gives at most 4 bins and the mode breaches the 10% ...
EarnSupBig	pre-screen checks	mass point by construction: every non-big firm in an FF48 industry-month receives the same value (the big firms' mean surprise), so each ...
ExchSwitch	pre-screen checks	binary exchange-switch flag: buildable from ACTIONS exchangeto/exchangefrom (TICK-ERS.exchange is current-only), but it fires on 0.50% of universe ...
FirmAgeMom	pre-screen checks	defined only in the youngest age quintile (OSAP), so coverage cannot exceed ~20%: 18.3-20.0% of the universe in the 205 months where the quintile is ...

predictor	class	reason recorded at the inventory
Frontier	pre-screen checks	coverage: OSAP does not zero-fill xrd, so rnd non-reporters and rnd == 0 (67% of non-null rnd in the universe) drop out; with the faithful filters ...
IndIPO	pre-screen checks	binary recent-IPO flag: 89.9% of the universe is 0 on average (78.8-95.4%), 2 distinct values, qcut 1 bin at all three probes; the modal share ...
IndMom	pre-screen checks	mass point by construction: the 2-digit-SIC cap-weighted past return is assigned to every industry member, so each cross-section holds 63-68 distinct ...
IndRetBig	pre-screen checks	mass point and coverage by construction: the big firms' FF48 mean return is assigned to the non-big members and the big firms are NaN; coverage ...
MomRev	pre-screen checks	two-valued flag (1 = top Mom6m and bottom Mom36m quintile, 0 = the reverse, NaN otherwise): coverage 4.2-16.7% of the universe (mean 9.0%, pooled ...
MomVol	pre-screen checks	categorical label 1..10 scored only in the top volume tercile: coverage 26.5-33.0% of the universe (ceiling 33.3%), under 40% in all 265 scorable ...
MS	pre-screen checks	six-valued score (MS in 1..6): 6 distinct values in every measured month, modal share 20.8-52.9% (mean 25.0%), >= 10% cliff in all 269 measured ...
NumEarnIncrease	pre-screen checks	integer count 0..8 of consecutive quarterly earnings increases: 9 distinct values, modal value 0 at 34.9-84.6% of scored names (median 52.1%), >= 10% ...
OScore	pre-screen checks	OSAP's published predictor is a binary flag (1 = top O-Score decile, 0 = deciles 1-7, deciles 8-9 dropped): 2 distinct values, 87.3-87.5% of scored ...
PS	pre-screen checks	Piotroski F-score, an integer 0..9 scored in OSAP's top book-to-market quintile: 6-9 distinct values, modal share 20.0-40.7% (median 26.8%) >= 10% in ...
RD	pre-screen checks	coverage: OSAP does not zero-fill xrd and scores only R&D reporters; SF1.rnd is vendor-zero for non-reporters, so rnd == 0 is NaN (faithful reading): ...
RDAbility	pre-screen checks	coverage by construction: scored only in the top tercile of R&D/sales with R&D > 0 (6.9% of the universe) and after 8 fiscal-year rows with >= 6 ...
RDCap	pre-screen checks	OSAP keeps only the bottom market-cap tercile of all listed stocks, which contains 0 harness-universe names in all 276 months (coverage 0%); and ...
RDIPO	pre-screen checks	binary flag (recent IPO with zero R&D): modal zero share 92.5-98.4% in every month, qcut 1 bin; also approx IPO dates (TICKERS.firstpricedate floored ...
ShareRepurchase	pre-screen checks	binary repurchase flag (prstk > 0): 2 distinct values, modal share 50.0-72.3% every month, qcut 1-2 bins; also approx data (gross prstk not in ...
ShareVol	pre-screen checks	OSAP's published signal is {0,1} at 5% and 10% share-turnover thresholds with the middle dropped: under the plain-percent reading 95.3% of names are ...
Spinoff	pre-screen checks	binary spin-off child flag (ACTIONS spunofffrom, 24-month window): 1.06% of the universe flagged on average (20.5 names a month; < 30 in 237 of 276 ...
std_turn	pre-screen checks	OSAP nulls std_turn for market-cap quintiles 4-5 of all listed stocks; the harness universe sits almost entirely above the 60th percentile, so ...
SurpriseRD	pre-screen checks	binary R&D-surprise flag: 2 distinct values, qcut 1 bin in >= 272 of 276 months under either reading; with rnd == 0 as missing (OSAP does not ...
DelBreadth	data start	constructible from 13F (SF3 investor-level or SF3A shrholders) but 13F starts 2013-06-30: 97 decision months with a 45-day point-in-time lag ...
FirmAge	data start	age = months since first trade is censored at the dataset start (SEP and TICKERS.firstpricedate begin 1997-12; OSAP censors at 1926 and nulls ...
MomOffSeason11YrPlus	data start	data start: 55 off-season returns at lags 120..178 need the close at BME(t-179); with SEP from 1997-12 the first full-window signal is 2012-11-30, so ...
MomOffSeason16YrPlus	data start	data start: 55 off-season returns at lags 180..238 need the close at BME(t-239); first full-window signal 2017-11-30, so only 49 of 276 decision ...
MomSeason11YrPlus	data start	data start: 5 same-month returns at lags 131..179 need the close at BME(t-180); first full-window signal 2012-12-31, so only 108 of 276 decision ...
MomSeason16YrPlus	data start	data start: 5 same-month returns at lags 191..239 need the close at BME(t-240); first full-window signal 2017-12-29, so only 48 of 276 decision ...
RIO_MB	data start	data start: 13F institutional ownership (SF3A) from 2013-06-30 gives 94 scorable decision months with the 45-day filing lag (95 without), under ...
RIO_Turnover	data start	data start: as RIO_MB, 94 scorable months with 13F data < 120; 5-valued output (modal share ~28.9%), ~19.6% coverage (top turnover quintile only)
RIO_Volatility	data start	data start: defined by 13F institutional ownership (Sharadar SF3/SF3A from 2013-06-30): 96 scorable decision months with OSAP timing, 94 with a ...

Appendix D. Agents, Artefacts and the Written Record

Table DI
Division of Labour between Agents and Mechanism

Each component of the loop, its role, and what it may and may not do.

Component	Role	What it may and may not do
Runner (Claude Opus)	Runs the loop from the standing instruction file: reads the state, launches sub-agents, runs the harness, writes the journal, commits and tags, locally.	May decide anything not on the stop-and-ask list. May not edit a past result, move a stamp while a run is unevaluated, push or add a remote, or read anything outside the project.
Advisor (Claude Fable)	A stronger model consulted at every phase boundary, before every acceptance is committed, before any stop-and-ask is raised, and whenever a result looks too good.	Advises only; advice that changes a decision is logged.
osap-fetcher	Fetches one predictor’s reference code and catalogue row at the pinned commit and writes a construction-only specification with its Sharadar field mappings.	May never pull from the catalogue’s main branch. The specification carries no verdict from anywhere.
sharadar-field-checker	Verifies that every field a factor will use exists in the dataset with the expected meaning, units and null semantics, by querying the data files.	Writes the field map; one writer at a time. A field that exists but means something subtly different is the failure mode it exists for.
sharadar-translator	Turns a reviewed specification into one factor file and runs the pre-screen data checks on it.	May not write universe filters, dates, rebalance logic, sector ranking, hedging or statistics. Leaves the family unset.
alpha-reviewer	Adversarial audit of a factor or harness change for look-ahead, survivorship and point-in-time errors.	Read-only. Invoked after every translated batch, after any change to the data layer, and whenever a result looks too good.
factor-evaluator	Parses a completed run, checks the four stamps against the repository, applies the bars, explains the numbers, writes every record; on an acceptance, implements the new version.	Refuses a mismatched stamp. Its verdict must equal the stamped one; a disagreement is a harness defect. Reports the construction layer and never judges it.
Records checker and hooks	Rebuilds the registry index, enforces the phase checkpoints, reconciles the list of untested predictors against the catalogue, refuses the vendor data files, credentials, an unproven harness change, an incomplete results file and any edit to a committed result.	Enforced by code, not by conduct.
Written record	One registry row per candidate; events.jsonl (append-only); MODEL_MANIFEST.yaml (one entry per version, never edited); the decisions file; the journal; every run’s report and results; one git tag per version.	Past results are never edited; a correction is a new event or entry that says what it corrects.
Test suite	476 tests over the data layer, analytics, portfolio construction, the runner, the family blend, the within-sector ranks, the market hedge and the construction layer.	Runs before every commit that touches the harness, a factor or the configuration; a change without a passing run is refused by the hook.

Appendix E. Reproducibility

Table EI
The Repository at the Out-of-Sample Test

The state of the repository when the holdout was evaluated (run 054’s stamps). Every composite version has one annotated tag, from v0-baseline to v14-add-TrendFactor; the v1 tag annotates the commit before v1.

Stamp at the out-of-sample test	Value
HARNESS_SHA	1271266472a9
CONFIG_SHA	0d88328d5b10 (never moved by a run)
COMPOSITE_SHA	7fe6f001e708 (v14, tag v14-add-TrendFactor)
DATA_SHA	42587e08609a (moved once, from 198b281de1a0, by the D8 data update)

Stamp at the out-of-sample test	Value
UNIVERSE_SHA	1c6512b922a8
LAYER_SHA	4b279fc317cd
OSAP source	github.com/OpenSourceAP/CrossSection at commit b4e911e69678a7424f318617a61d813f54183123

Table EII
Reproduction Proofs

Each harness or data move that preceded a later number was first shown to reproduce the live composite. Sources: events phase_completed D, docs/JOURNAL.md (Phase E entries), the results of the runs named.

run	against	what was compared	result
013, 015, ..., 042: each version's own baseline	the with-candidate model tested on the rung that accepted its last leg	every compared statistic, to six decimal places	66/66 on every version
045 (Stage 2) under HARNESS 471f70782486	042 (v14 Stage 2)	every statistic	equal except harness_sha
047, 048 (Stage 2, Stage 3) under HARNESS 3561590b660a	042, 043	every statistic	equal except harness_sha
052 (Stage 2) under the rf diagnostic harness	042	every statistic the old harness prints	equal; run 052 adds the excess-of-rf statistics
053 (Stage 2) on DATA 42587e08609a	052 on DATA 198b281de1a0	the D8 restatement	28 statistics identical, 54 re-stated; IC moved by -0.000009
054, in-sample portion	053	every shared statistic	20 of 20 equal to six decimal places
057 equal-rank deciles at \$100M, hold-out portion, gross	054 holdout portion, raw long-short	annual return	3.01 against 3.01

Rebuilding the paper's tables and figures reads records only: `python3 paper/build_tables.py` writes the source tables and every scalar with its source; `python3 paper/manuscript/build_sample.py` describes the universe from the frozen dataset with the harness's own screens; `python3 paper/manuscript/build_figures.py` draws the figures; and `python3 paper/manuscript/build_manuscript.py` assembles this document and lists every manuscript fact with its record source. Re-running a measurement needs the dataset, which is not redistributable, and the harness at the stamped commit: `python3 harness/run_test.py --baseline --stage 2` reproduces v14's in-sample results. The holdout has been used; re-running its commands on this dataset would mean examining it a second time.

Appendix F. The Accepted Legs, Family by Family

This appendix describes the fourteen accepted signals in Table IX, grouped by the family each one joined. Figures are the Stage 1 and Stage 2 registry rows, on the original data.

Profitability. Four legs joined the Fama-French operating-profitability characteristic, the most of any family, so each holds a forty-fifth of the composite. CBOperProf, the cash-based operating profitability of Ball, Gerakos, Linnainmaa, and Nikolaev (2016), strips accruals out of operating profit and scales by assets. Its IC of 0.0209 (t-stat 3.74) is uniform across the liquidity tiers, and its spread is a short-leg effect: it identifies cash-unprofitable firms to avoid rather than grading the winners. Financials and real estate are nearly absent by the catalogue's own exclusion. Its residual t-stat against v1 was 3.02, with a residual share of 0.50. GP, the gross profitability of Novy-Marx (2013), has the cleanest standalone record of the profitability passers: a raw Sharpe of 0.688, a hedged Sharpe without its three best years of 0.92, and a hedged annual long-short negative in only two years. It cleared at 2.74. roaq, the quarterly return on assets of Balakrishnan, Bartov, and Faurel (2010), is a quarterly signal whose IC halves after a quarter; its residual of 3.73 carried it against a base that already held three profitability legs: the seed, CBOperProf and GP. RoE, net income over book equity in the catalogue's Haugen and Baker (1996) construction, shares its denominator with the seed leg and cleared at 2.41 with the lowest residual share of its ladder. OperProfRD was rejected in this family, its information already held.

Investment. PctAcc, the percent accruals of Hafzalla, Lundholm, and Van Winkle (2011), joined the asset-growth characteristic under the decision that absorbed the accruals label into investment. Its IC is modest at 0.0112 but steady (t-stat 4.03), its raw spread has the best Sharpe of the passers at 0.762, and its beta is near zero. It was the first rung of the ratchet, at a residual t-stat of 4.67 and a residual share of 0.88, the information showing in the composite's IC rather than in its spread (Section II.F).

External financing. This family was opened by ShareIss5Y, Daniel and Titman's (2006) five-year share issuance, and filled by XFIN, the net external financing of Bradshaw, Richardson, and Sloan (2006). ShareIss5Y is the lowest-turnover signal among the passers, at under seven

percent a month, with an IC nearly flat over horizons and the strongest guard statistic of the search (+2.83); it scores from June 2003 because of the 65-month history it needs. XFIN, the sum of net equity and net debt financing over assets, cleared the residual bar at 3.27 while every diagnostic of its rung moved against it, the case Section VI describes. ShareIss1Y and NetEquityFinance, whose annual IC series move 0.95 with XFIN's, were rejected as spanned.

Value. cfp, operating cash flow to price from Desai, Rajgopal, and Venkatachalam (2004), joined book-to-market. Its IC of 0.0246 (t-stat 3.65) is front-loaded, 0.039 in the first half against 0.010 in the second, and two thirds of its summed hedged long-short sits in 2000, 2002 and 2021. Its stock-level residual of 2.97 and its portfolio spanning alpha (t-stat 0.40) disagree, and both are reported. CF, whose annual IC moves 0.96 with cfp's, and NetPayoutYield were rejected in this family.

Momentum. TrendFactor, from Han, Zhou, and Zhu (2016), forecasts next month's return from moving averages of price at several horizons, with the weights on those averages estimated on trailing cross-sections. It is almost uncorrelated with the composite (a spanning R^2 of 0.01), its standalone raw return is the largest of the passers at 10.1% a year, and its beta is near zero. Its IC one month on is close to zero, its turnover is 63% a month, and one year, 2009, carries half of its summed hedged long-short. It scores from 2003 and cleared at 2.14.

Volatility. MaxRet, the maximum daily return in the signal month from Bali, Cakici, and Whitelaw (2011), opened the volatility family at v7. It has one of the largest standalone ICs of the passers, 0.0233 (t-stat 3.42), uniform across tiers, and a residual t-stat of 4.01. Its portfolio is weak and short the market: a raw Sharpe of 0.192, a beta of -0.86 , and turnover of 71% a month. Its rung is the clearest case of the hedge reaching the guard (Section VI). IdioVol3F, the one-month idiosyncratic volatility of Ang, Hodrick, Xing, and Zhang (2006), joined it at the thinnest margin of the search, 2.03; its beta of -0.98 is the most negative of the accepted legs. RealizedVol and IdioVolAHT were rejected in this family.

Short-term reversal. STreversal, last month's return from Jegadeesh (1990), opened its own family, a ninth of the composite. It is the month the momentum seed skips, and its scores

barely correlate with momentum's. Its IC of 0.0145 (t-stat 2.96) lives in the small tier, it turns over 86% of its top decile a month, and its beta is positive, so the hedge turns its raw spread negative. Its residual exceeded its own IC, which triggered and passed an audit (Section VI).

Liquidity. `zerotrade6M`, the zero-trading-day and turnover measure of Liu (2006) over six months, opened the liquidity family at v12. In this universe almost no name has a zero-volume day, so the signal is in effect low turnover; its raw beta is -0.73 , and its annual IC moves with the volatility legs (Section X). It cleared at 2.69. `VolumeTrend`, from Haugen and Baker (1996), is the slope of a stock's share volume over the past five years scaled by its mean volume, with a falling trend attractive; it is a mid-tier effect with a thin IC margin, scores from 2003, and cleared at 2.04. The other two zero-trade windows and `BidAskSpreadFlip` were rejected in this family.

Size gained no leg; no Stage 1 passer carried the size label.